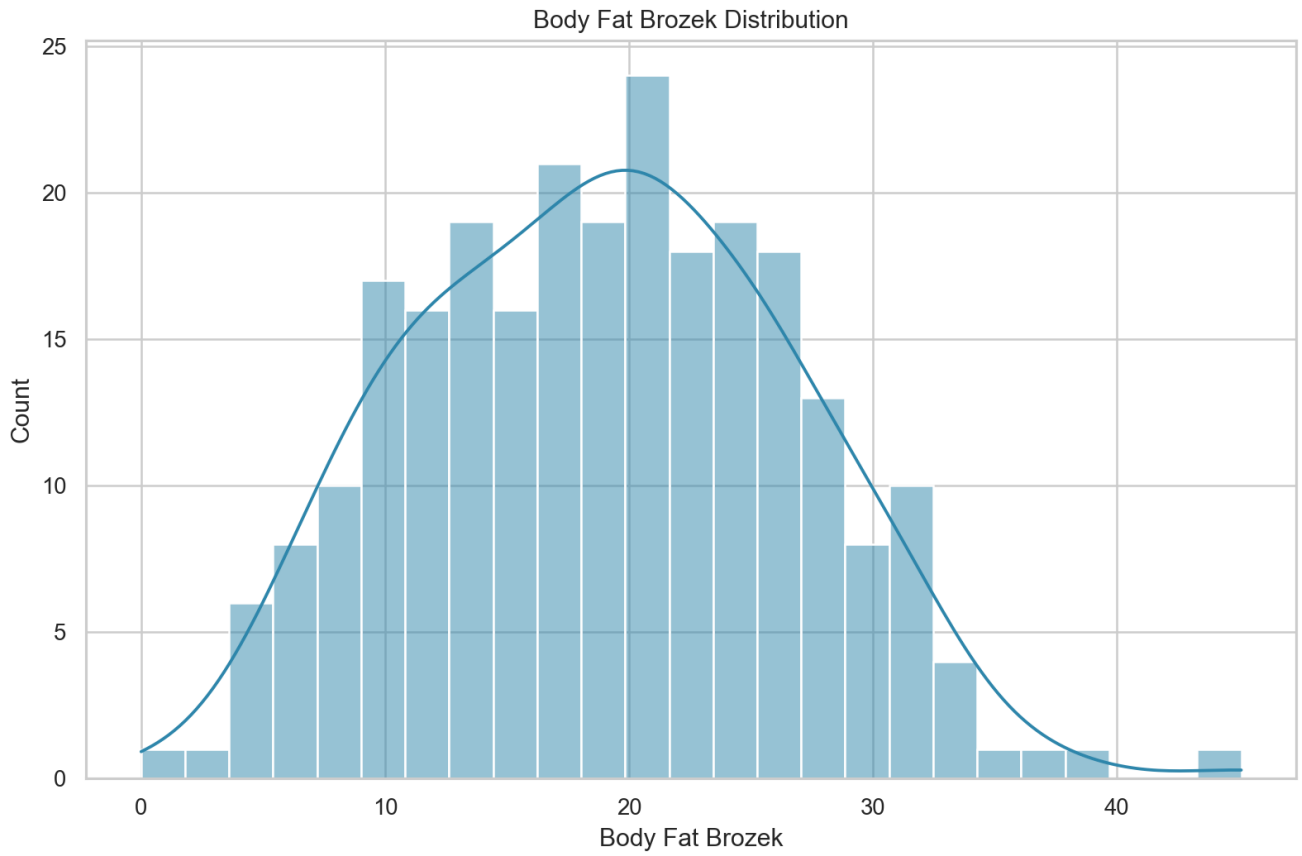
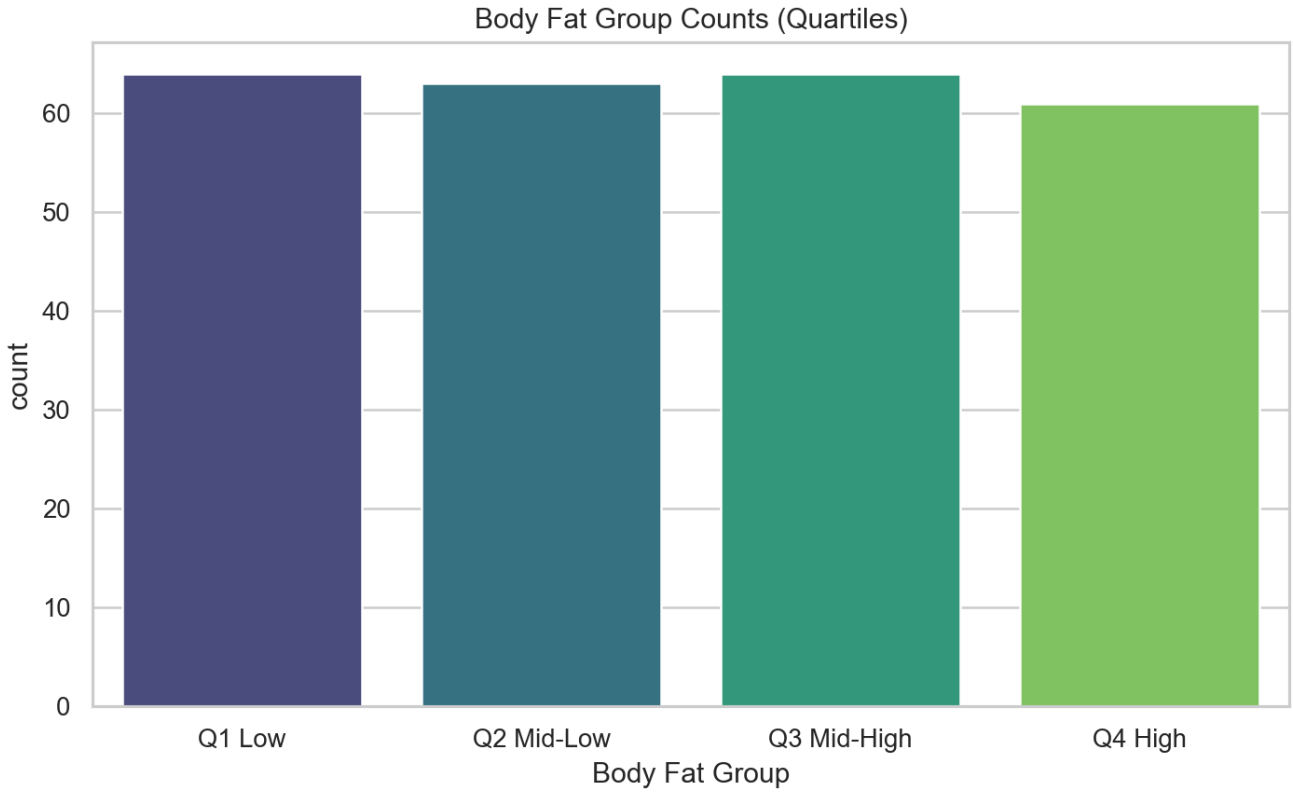
Body Fat Brozek 시각 분석 보고서 (미리보기 중심)

1) 데이터 개요

- 데이터 행 수: **252**
- 타겟 변수: `body fat brozek`
- 분석 목표: 분포, 이상치, 상관관계, 변수별 패턴을 직관적으로 확인

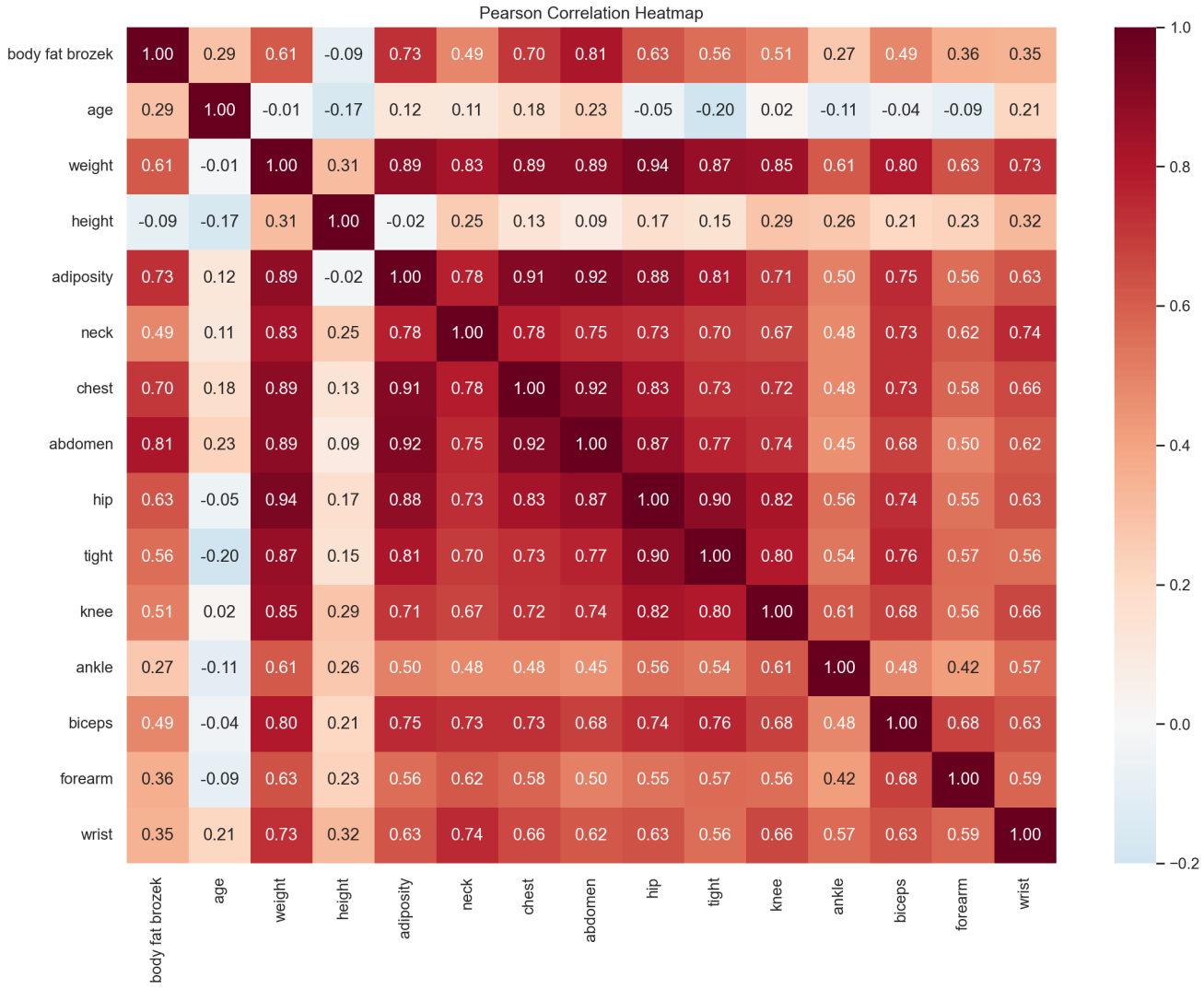
초기 분포를 보면 체지방 수치가 단봉형 분포를 가지며, 사분위 그룹은 거의 균등하게 나뉩니다.





2) 상관관계 중심 해석

Pearson/Spearman 히트맵에서 체지방과 함께 움직이는 핵심 변수는 `abdomen`, `adiposity`, `chest`, `hip`, `weight`입니다.


Pearson Correlation Heatmap
body fat brozek 1.00 0.29 0.61 -0.09 0.73 0.49 0.70 0.81 0.63 0.56 0.51 0.27 0.49 0.36 0.35
age 0.29 1.00 -0.01 -0.17 0.12 0.11 0.18 0.23 -0.05 -0.20 0.02 -0.11 -0.04 -0.09 0.21
weight 0.61 -0.01 1.00 0.31 0.89 0.83 0.89 0.89 0.94 0.87 0.85 0.61 0.80 0.63 0.73
height -0.09 -0.17 0.31 1.00 -0.02 0.25 0.13 0.09 0.17 0.15 0.29 0.26 0.21 0.23 0.32
adiposity 0.73 0.12 0.89 -0.02 1.00 0.78 0.91 0.92 0.88 0.81 0.71 0.50 0.75 0.56 0.63
neck 0.49 0.11 0.83 0.25 0.78 1.00 0.78 0.75 0.73 0.70 0.67 0.48 0.73 0.62 0.74
chest 0.70 0.18 0.89 0.13 0.91 0.78 1.00 0.92 0.83 0.73 0.72 0.48 0.73 0.58 0.66
abdomen 0.81 0.23 0.89 0.09 0.92 0.75 0.92 1.00 0.87 0.77 0.74 0.45 0.68 0.50 0.62
hip 0.63 -0.05 0.94 0.17 0.88 0.73 0.83 0.87 1.00 0.90 0.82 0.56 0.74 0.55 0.63
tight 0.56 -0.20 0.87 0.15 0.81 0.70 0.73 0.77 0.90 1.00 0.80 0.54 0.76 0.57 0.56
knee 0.51 0.02 0.85 0.29 0.71 0.67 0.72 0.74 0.82 0.80 1.00 0.61 0.68 0.56 0.66
ankle 0.27 -0.11 0.61 0.26 0.50 0.48 0.48 0.45 0.56 0.54 0.61 1.00 0.48 0.42 0.57
biceps 0.49 -0.04 0.80 0.21 0.75 0.73 0.73 0.68 0.74 0.76 0.68 0.48 1.00 0.68 0.63
forearm 0.36 -0.09 0.63 0.23 0.56 0.62 0.58 0.50 0.55 0.57 0.56 0.42 0.68 1.00 0.59
wrist 0.35 0.21 0.73 0.32 0.63 0.74 0.66 0.62 0.63 0.56 0.66 0.57 0.63 0.59 1.00
body fat brozek
age
weight
height
adiposity
neck
chest
abdomen
hip
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knee
ankle
biceps
forearm
wrist
1.0
0.8
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0.0
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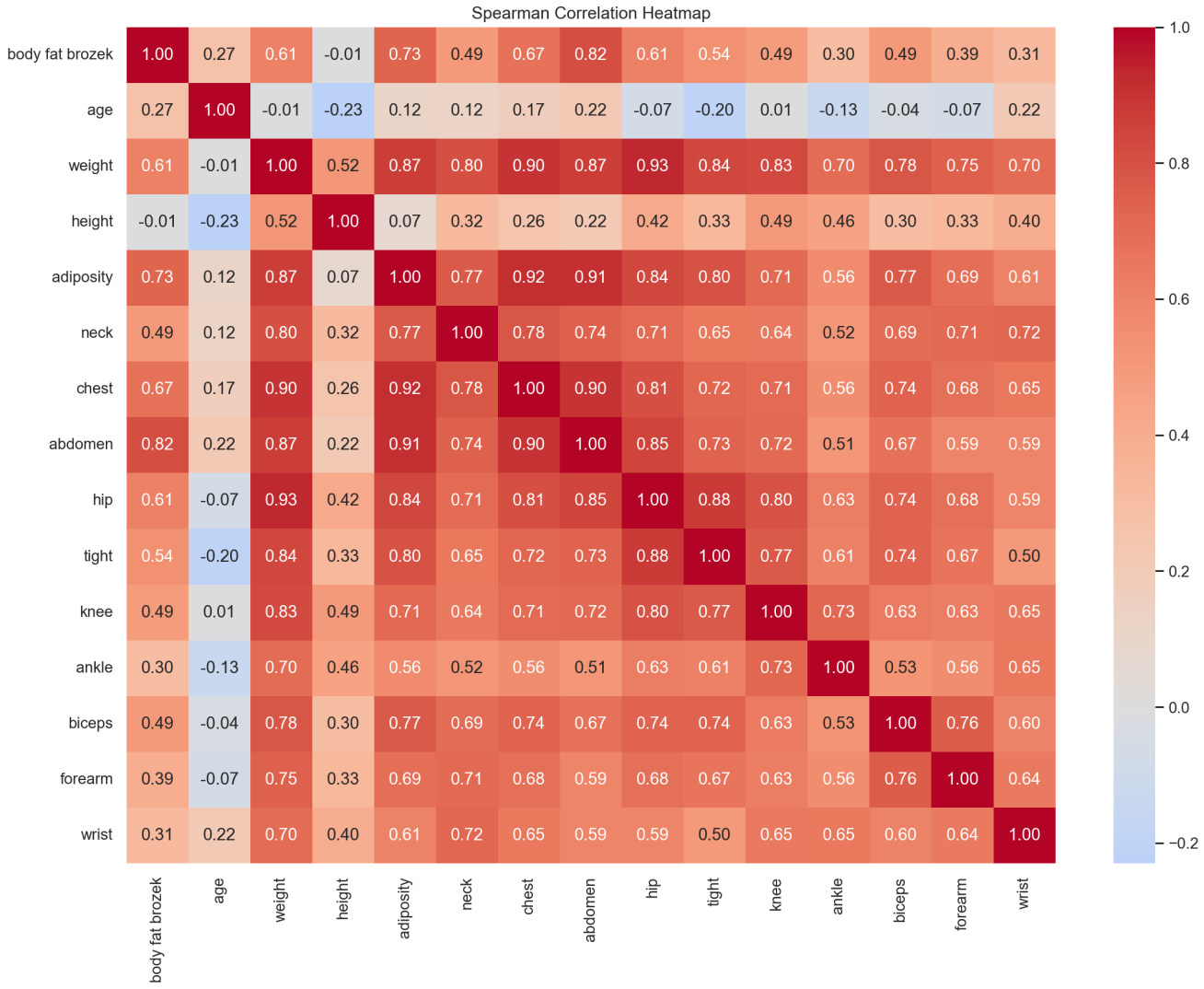
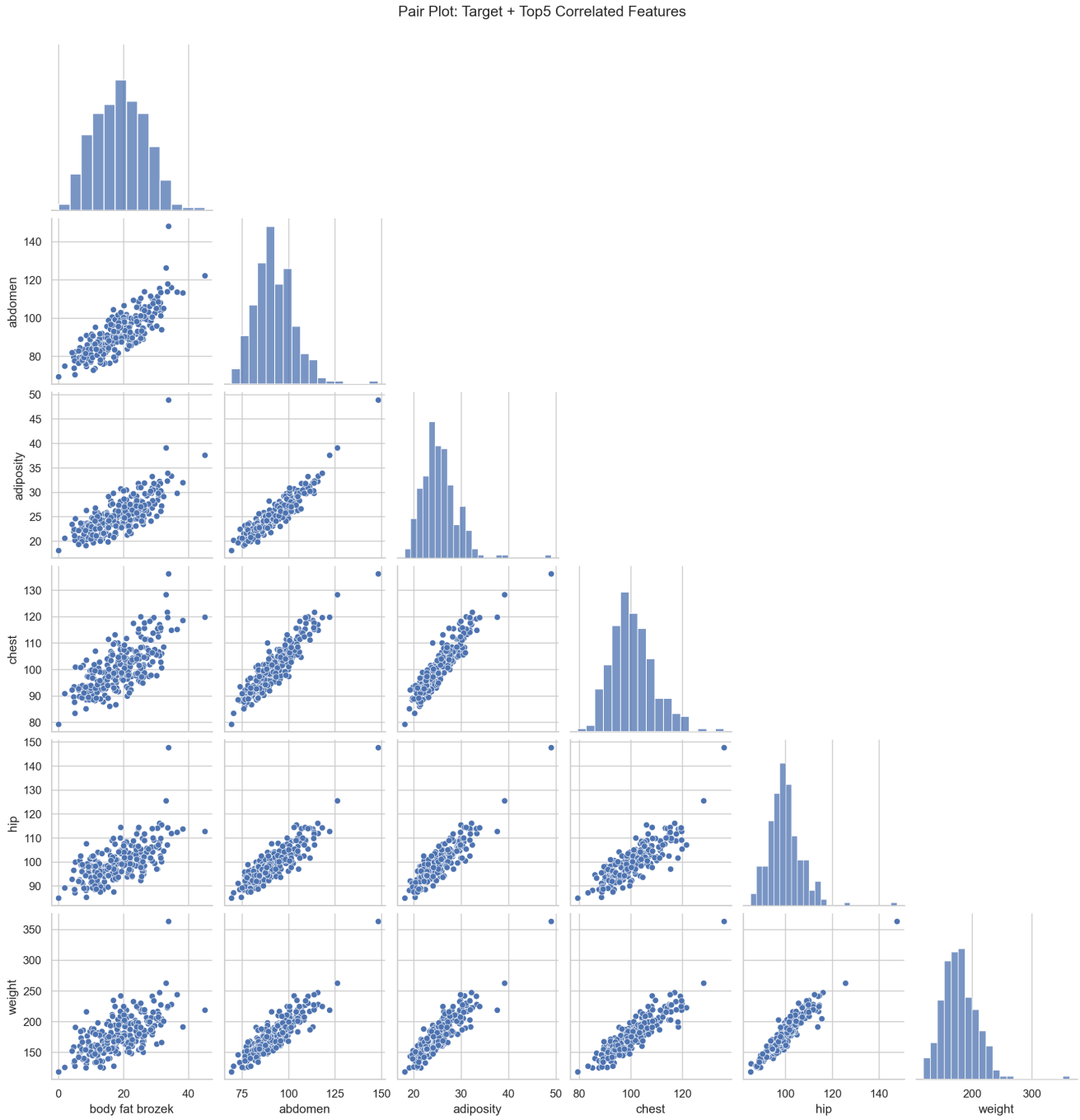

Spearman Correlation Heatmap
body fat brozek 1.00 0.27 0.61 -0.01 0.73 0.49 0.67 0.82 0.61 0.54 0.49 0.30 0.49 0.39 0.31
age 0.27 1.00 -0.01 -0.23 0.12 0.12 0.17 0.22 -0.07 -0.20 0.01 -0.13 -0.04 -0.07 0.22
weight 0.61 -0.01 1.00 0.52 0.87 0.80 0.90 0.87 0.93 0.84 0.83 0.70 0.78 0.75 0.70
height -0.01 -0.23 0.52 1.00 0.07 0.32 0.26 0.22 0.42 0.33 0.49 0.46 0.30 0.33 0.40
adiposity 0.73 0.12 0.87 0.07 1.00 0.77 0.92 0.91 0.84 0.80 0.71 0.56 0.77 0.69 0.61
neck 0.49 0.12 0.80 0.32 0.77 1.00 0.78 0.74 0.71 0.65 0.64 0.52 0.69 0.71 0.72
chest 0.67 0.17 0.90 0.26 0.92 0.78 1.00 0.90 0.81 0.72 0.71 0.56 0.74 0.68 0.65
abdomen 0.82 0.22 0.87 0.22 0.91 0.74 0.90 1.00 0.85 0.73 0.72 0.51 0.67 0.59 0.59
hip 0.61 -0.07 0.93 0.42 0.84 0.71 0.81 0.85 1.00 0.88 0.80 0.63 0.74 0.68 0.59
tight 0.54 -0.20 0.84 0.33 0.80 0.65 0.72 0.73 0.88 1.00 0.77 0.61 0.74 0.67 0.50
knee 0.49 0.01 0.83 0.49 0.71 0.64 0.71 0.72 0.80 0.77 1.00 0.73 0.63 0.63 0.65
ankle 0.30 -0.13 0.70 0.46 0.56 0.52 0.56 0.51 0.63 0.61 0.73 1.00 0.53 0.56 0.65
biceps 0.49 -0.04 0.78 0.30 0.77 0.69 0.74 0.67 0.74 0.74 0.63 0.53 1.00 0.76 0.60
forearm 0.39 -0.07 0.75 0.33 0.69 0.71 0.68 0.59 0.68 0.67 0.63 0.56 0.76 1.00 0.64
wrist 0.31 0.22 0.70 0.40 0.61 0.72 0.65 0.59 0.59 0.50 0.65 0.65 0.60 0.64 1.00
body fat brozek
age
weight
height
adiposity
neck
chest
abdomen
hip
tight
knee
ankle
biceps
forearm
wrist
1.0
0.8
0.6
0.4
0.2
0.0
−0.2



체지방과의 Pearson 상관 순위

상위 양의 상관:

	pearson_corr
abdomen	0.813706
adiposity	0.727994
chest	0.702885
hip	0.6257
weight	0.613156
tight	0.561284

중간 상관:

	pearson_corr
knee	0.507786
biceps	0.493031
neck	0.491489
forearm	0.363277

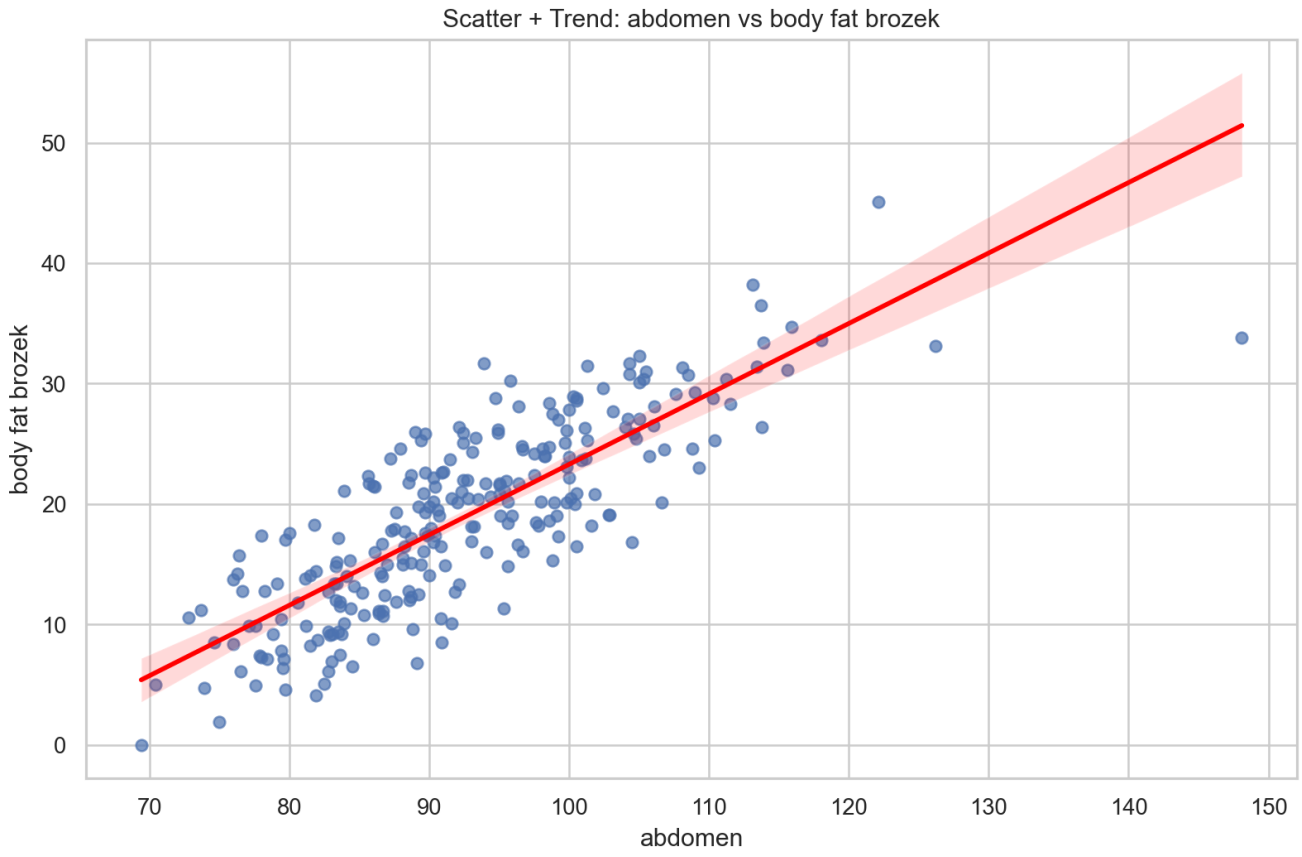
하위(음의 방향 포함):

	pearson_corr
wrist	0.347573
age	0.289174
ankle	0.266783
height	-0.0891064

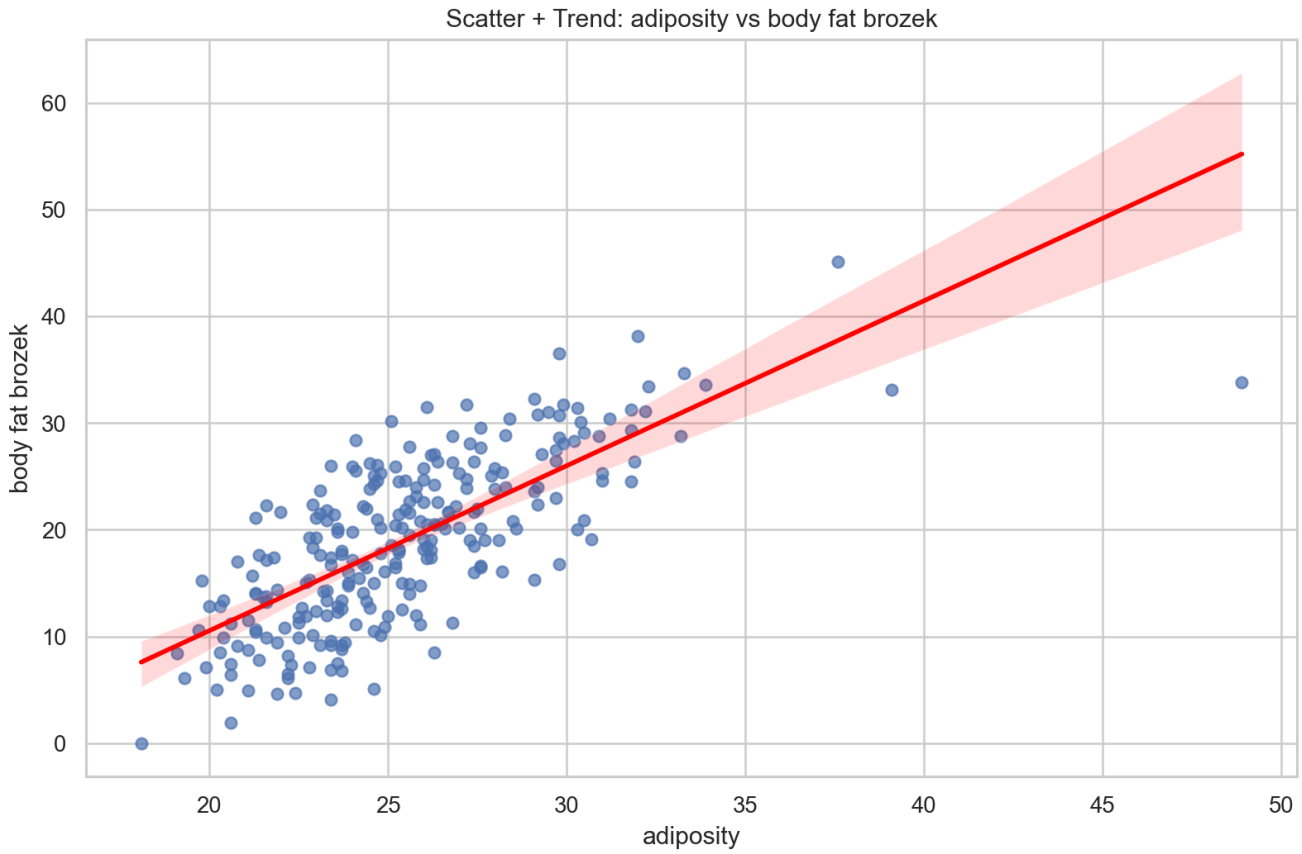
3) 핵심 변수별 산점도 + 추세선

체지방과의 관계를 직접 보는 섹션입니다. 복부(`abdomen`)는 가장 강한 양의 추세를 보이며, 키(`height`)는 약한 음의 경향을 보입니다.

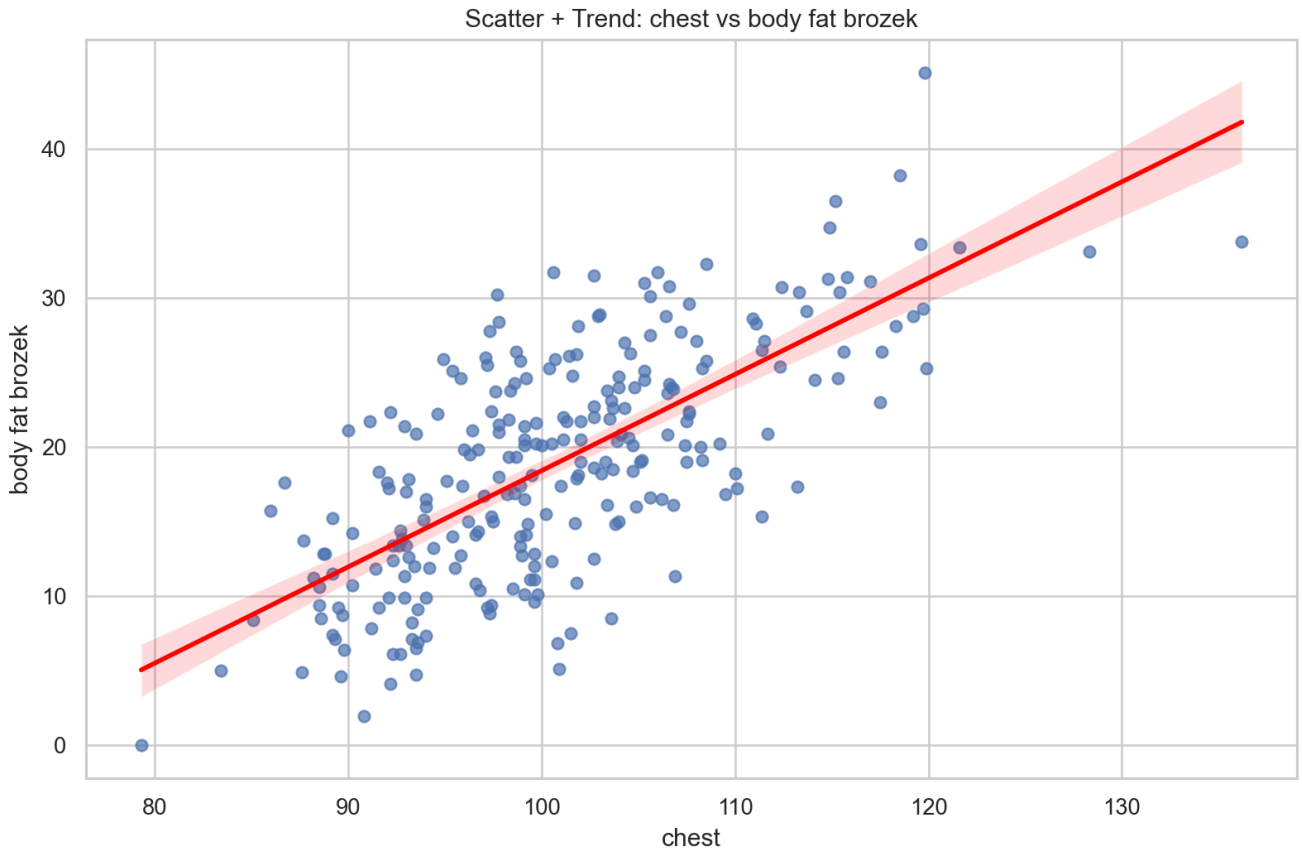
abdomen vs body fat



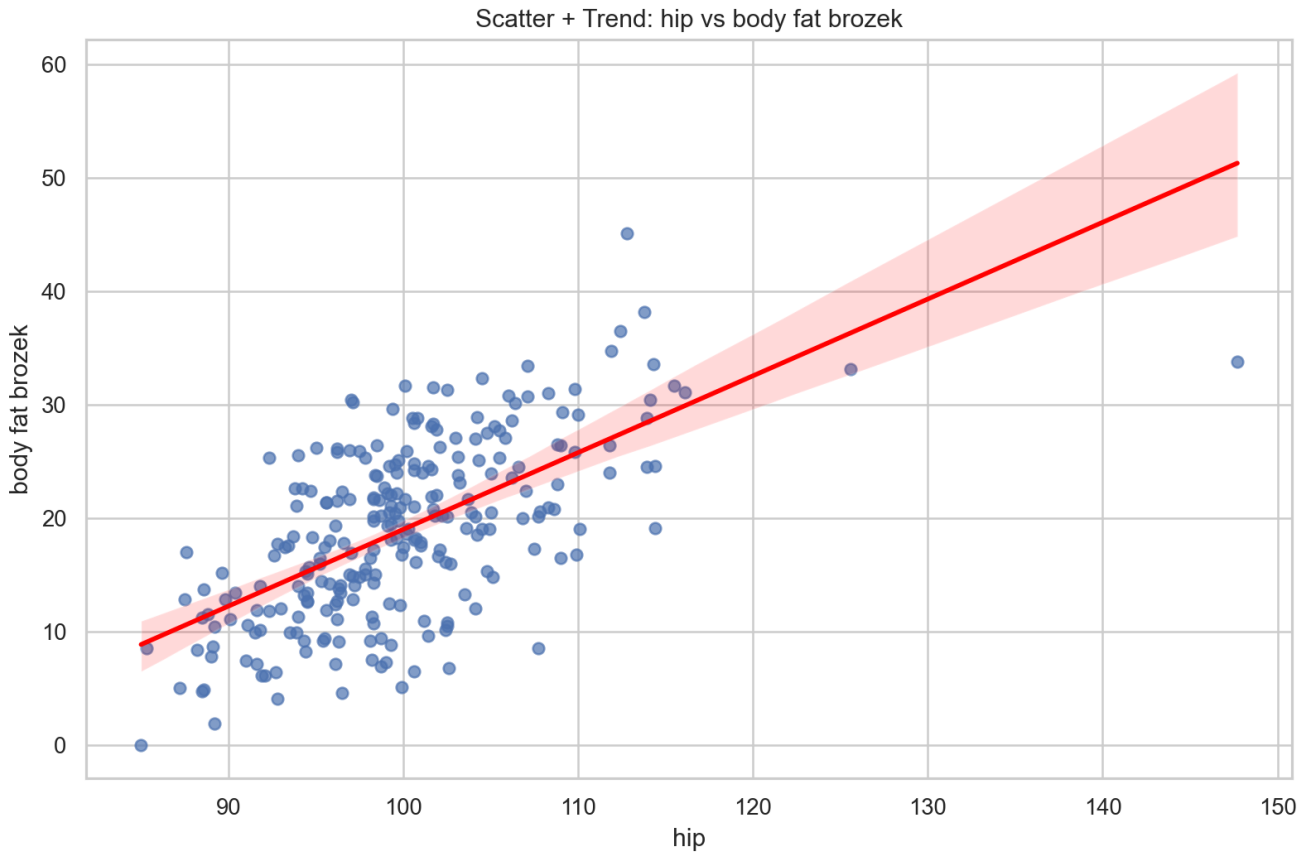
adiposity vs body fat



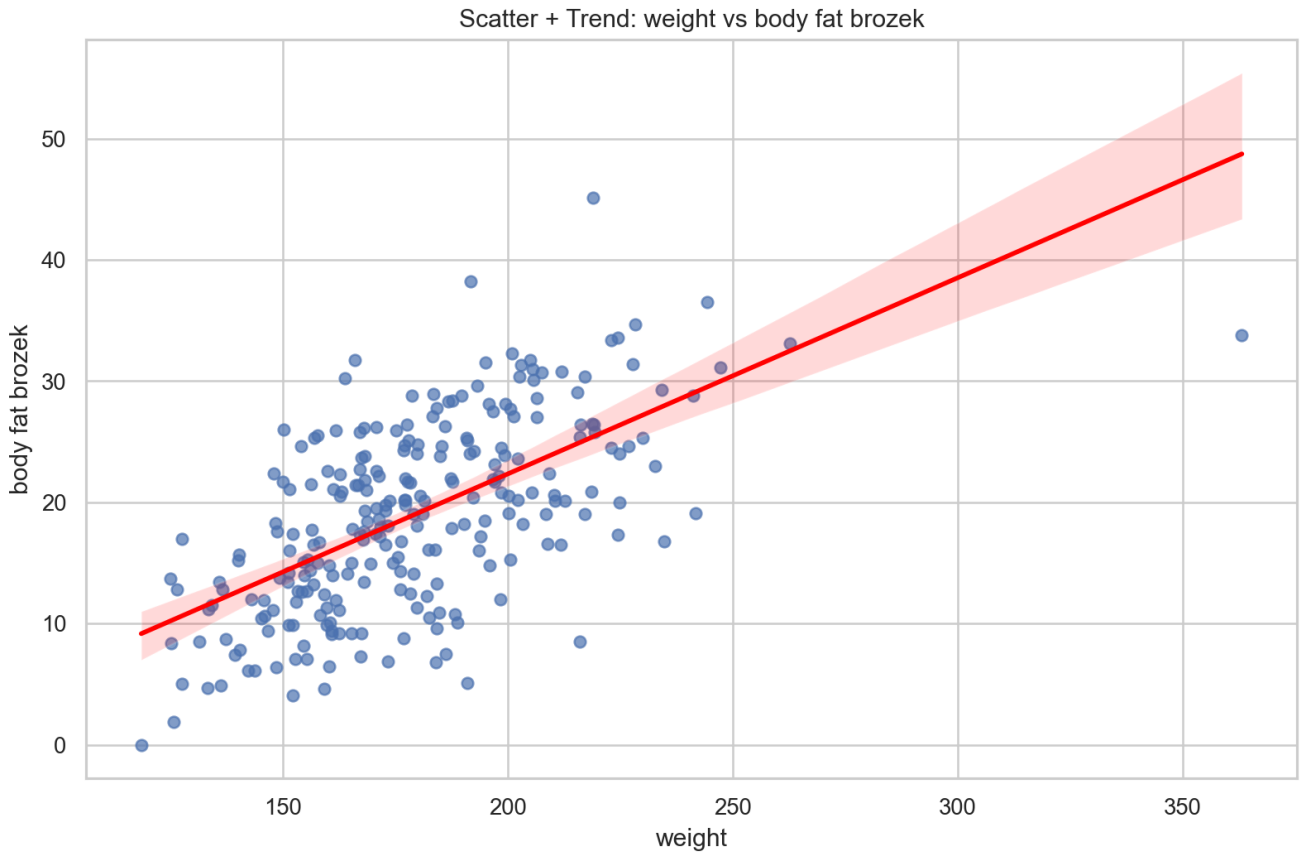
chest vs body fat



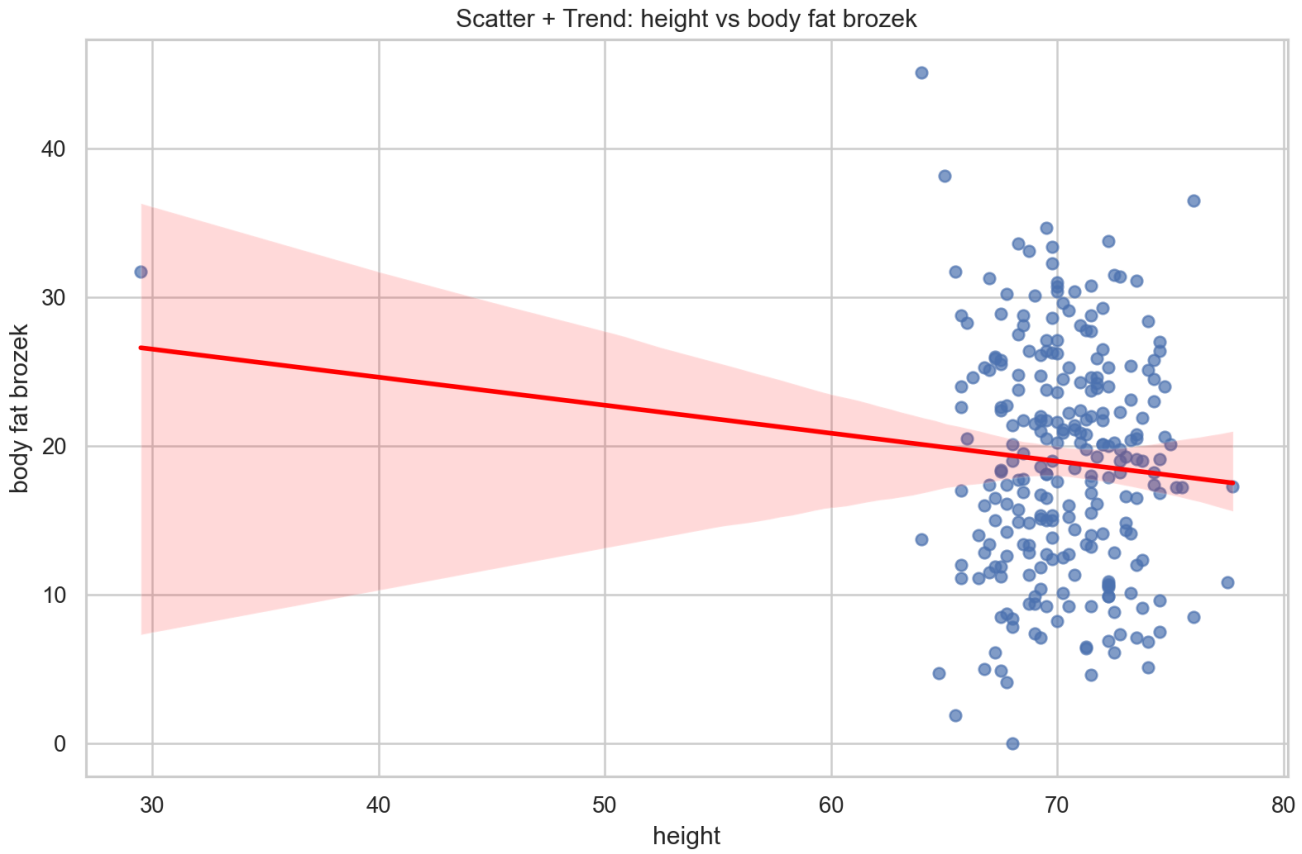
hip vs body fat



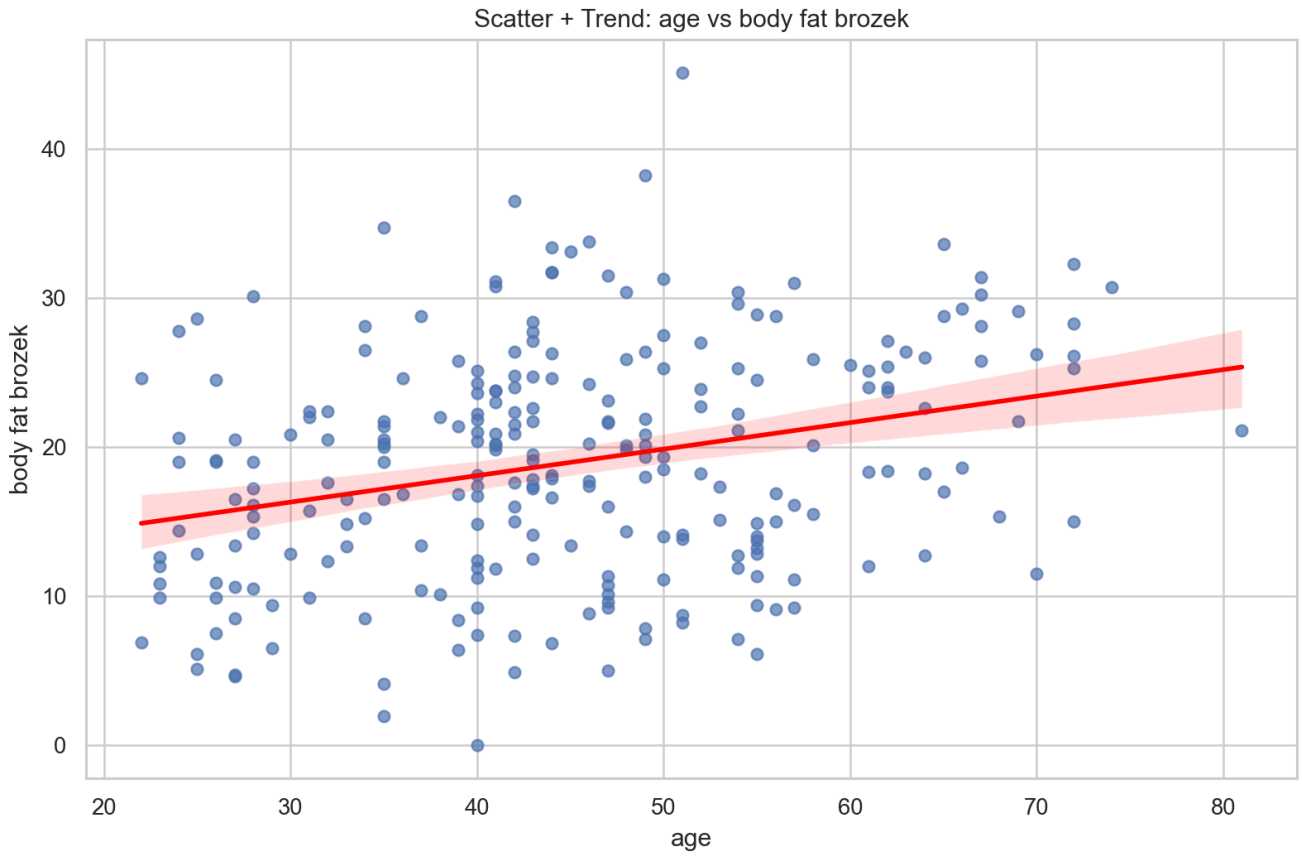
weight vs body fat



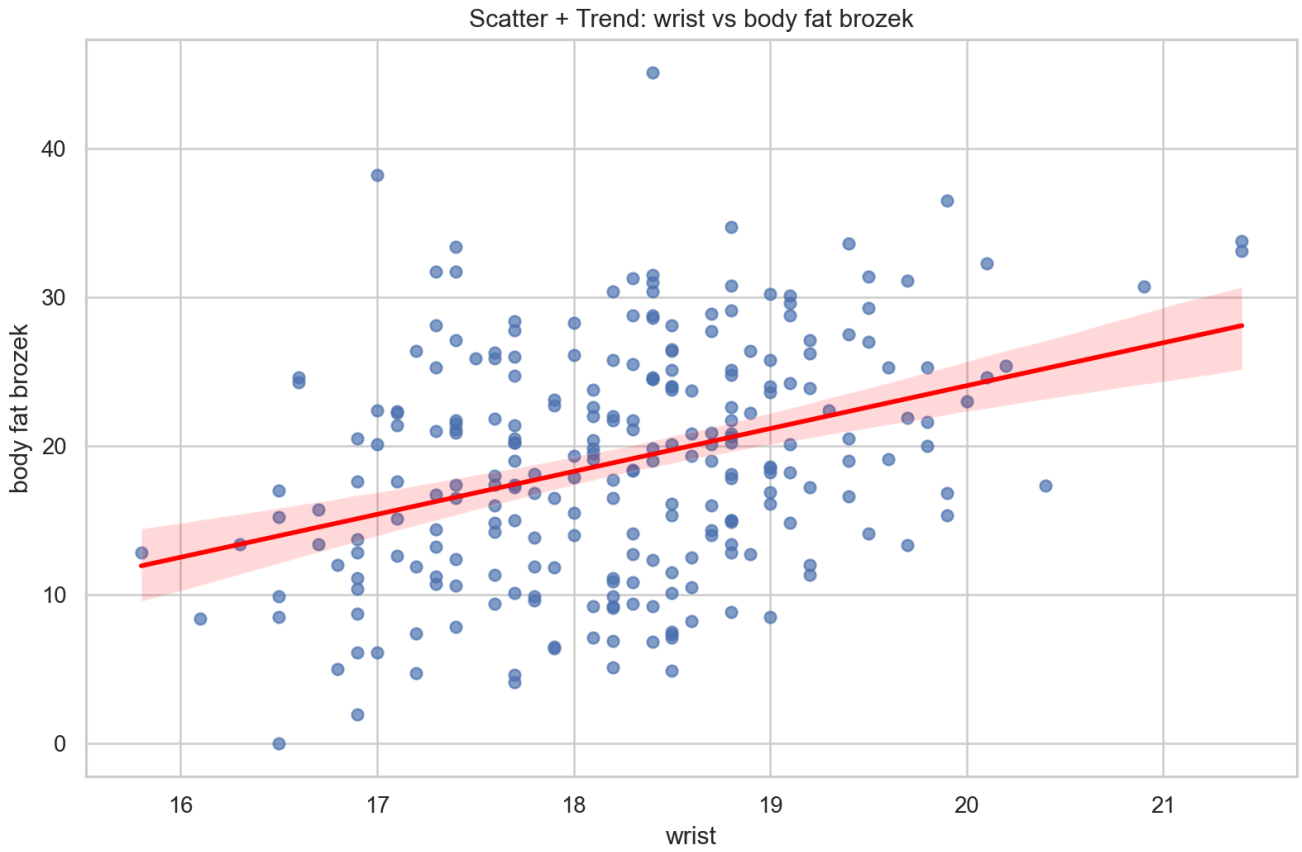
height vs body fat



age vs body fat



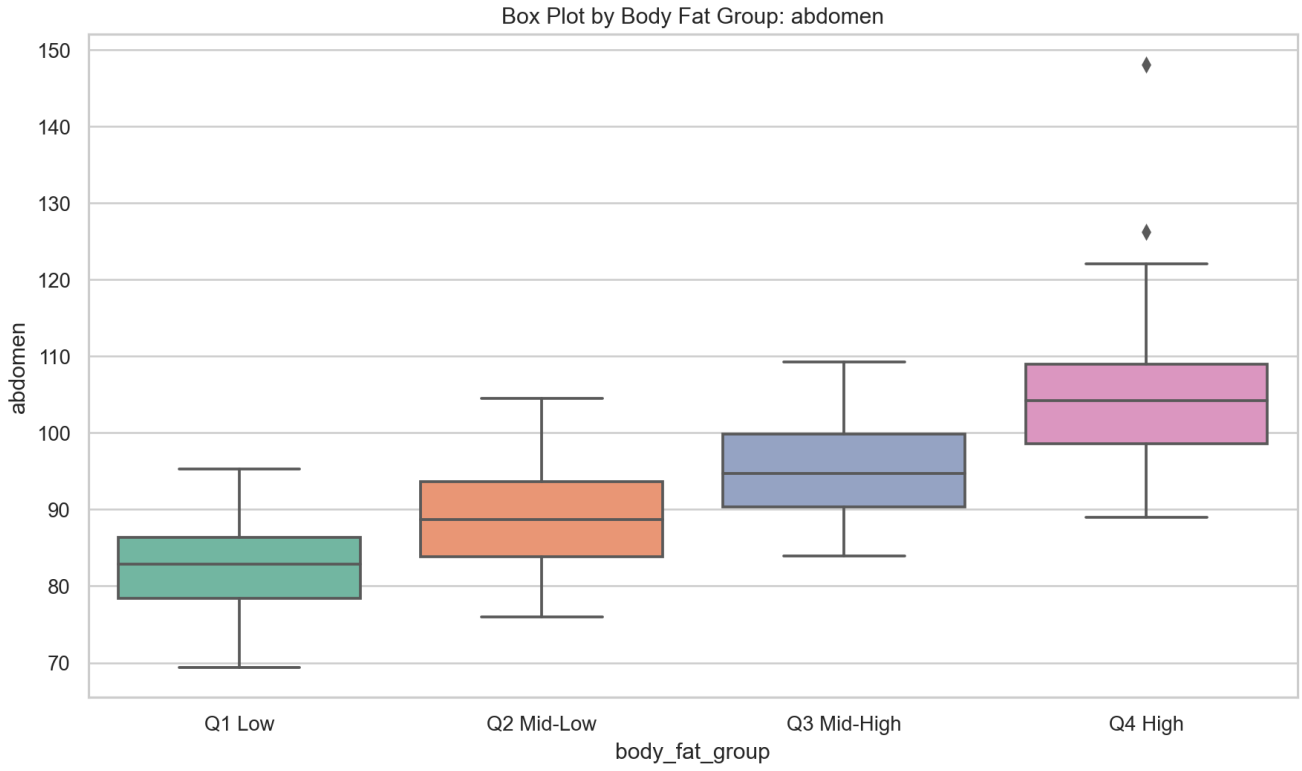
wrist vs body fat



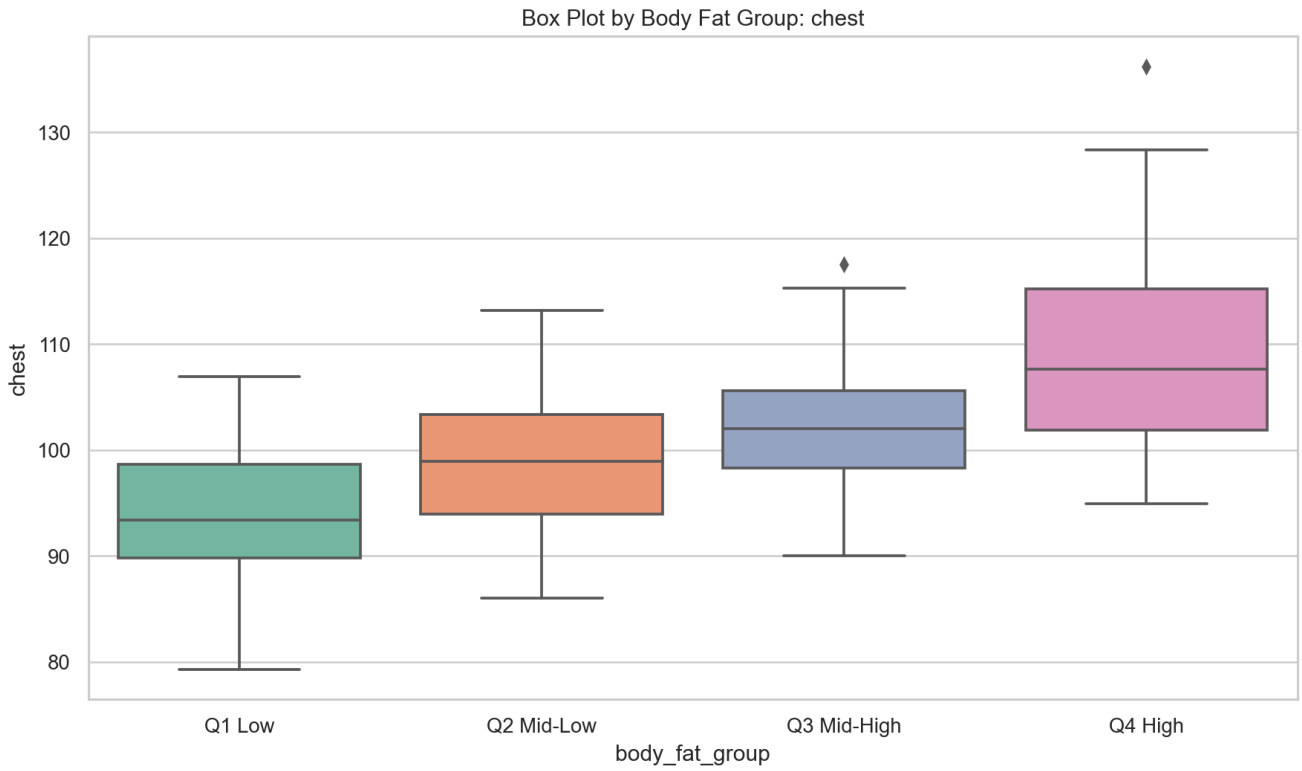
4) Box Plot 해석 (체지방 그룹별 분포)

Q1에서 Q4로 갈수록 다수 변수의 중앙값이 상승합니다. 특히 `abdomen`, `chest`, `hip`, `weight`에서 그룹 간 분리가 명확합니다.

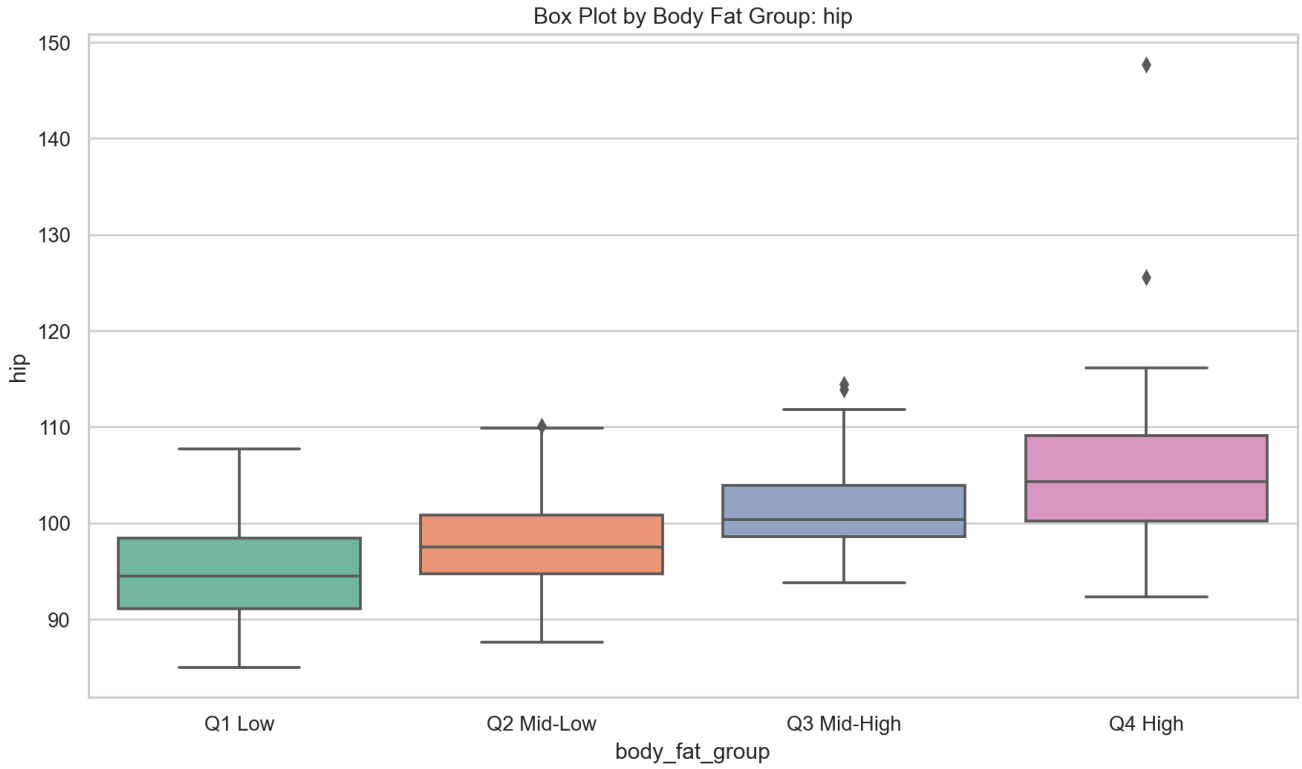
Grouped Box: abdomen



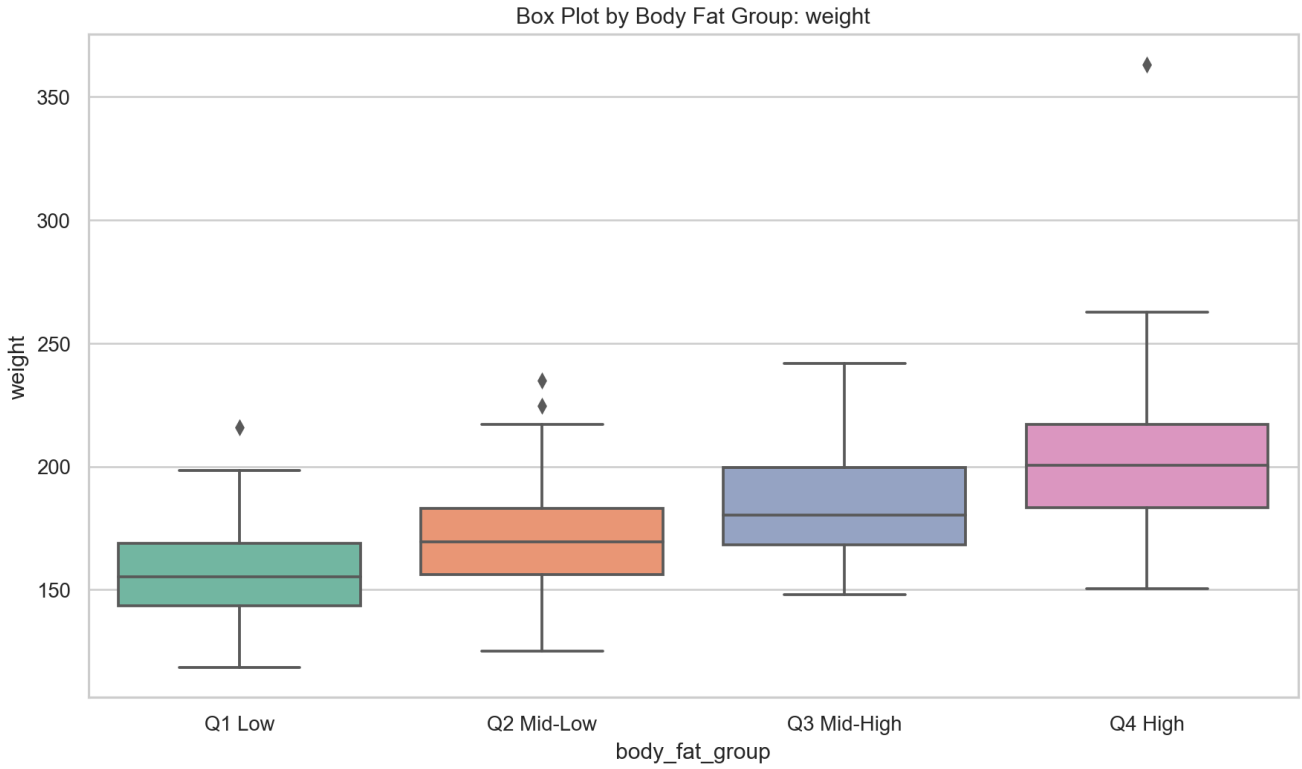
Grouped Box: chest



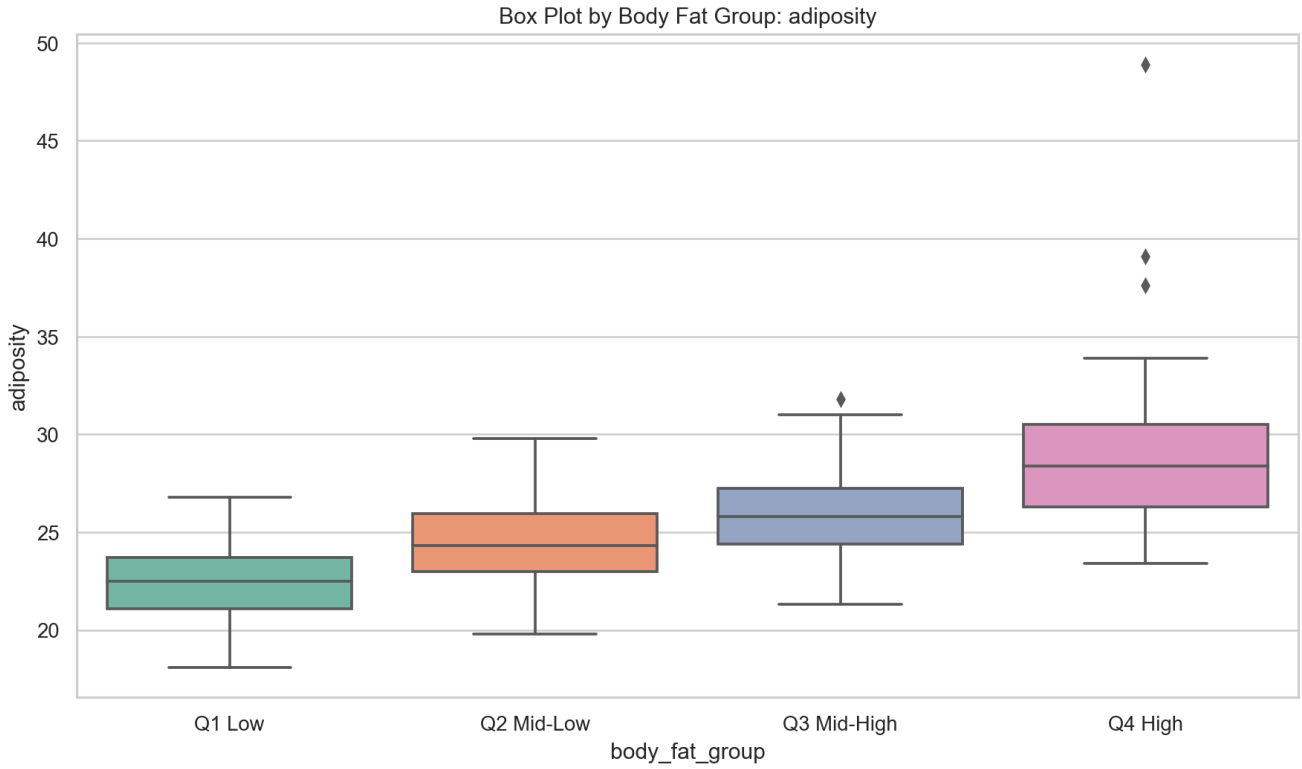
Grouped Box: hip



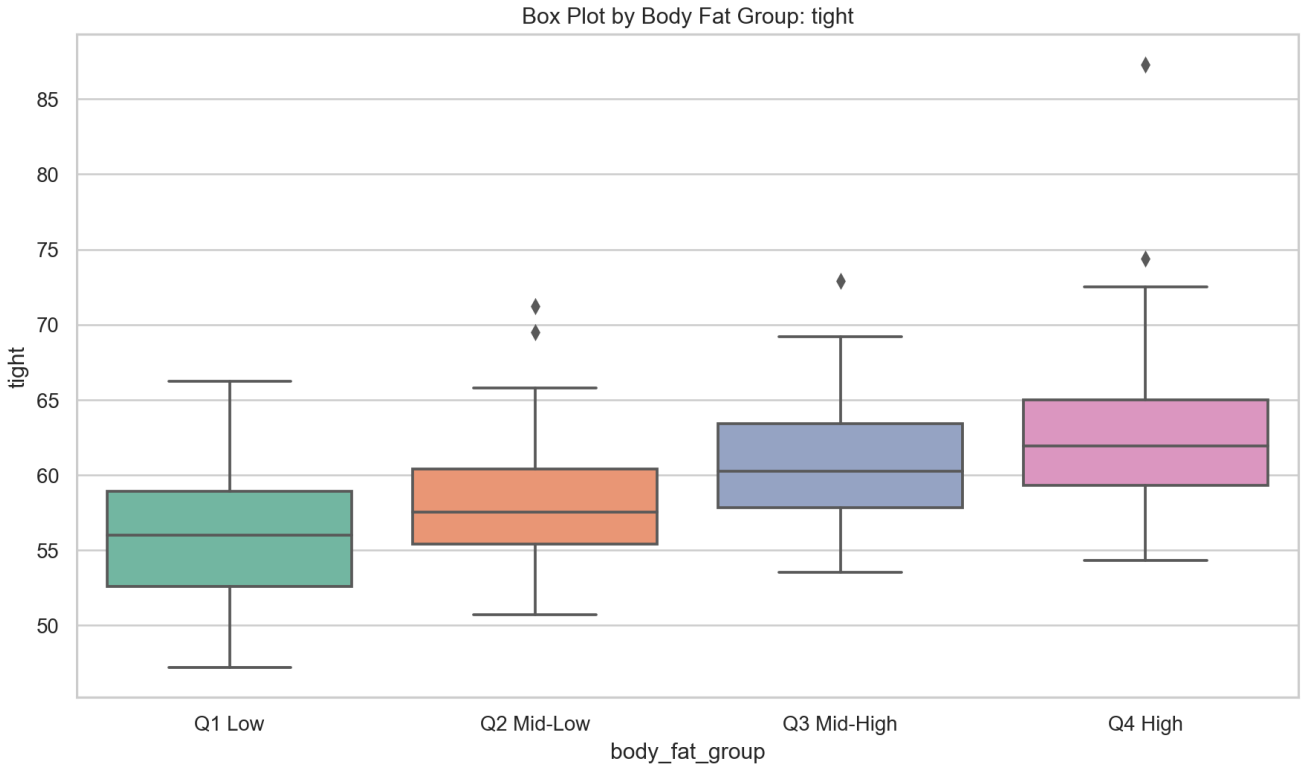
Grouped Box: weight



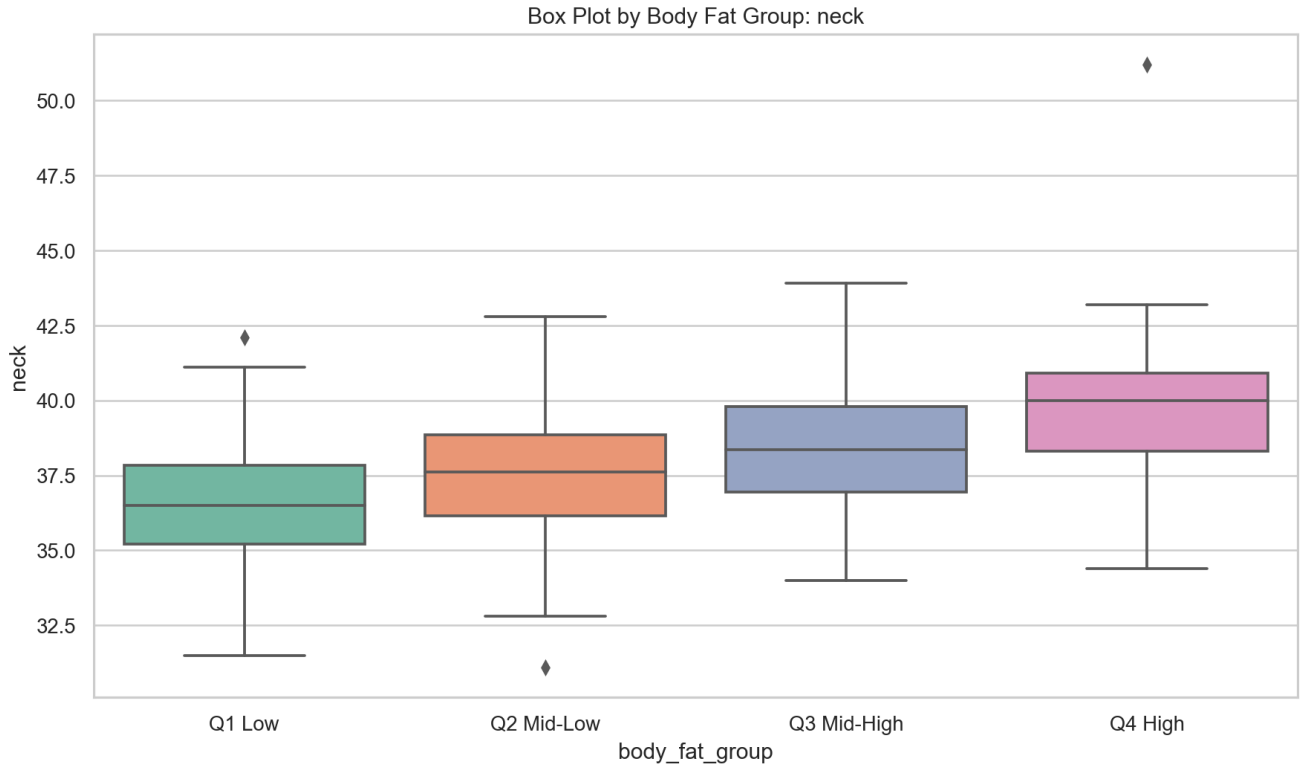
Grouped Box: adiposity



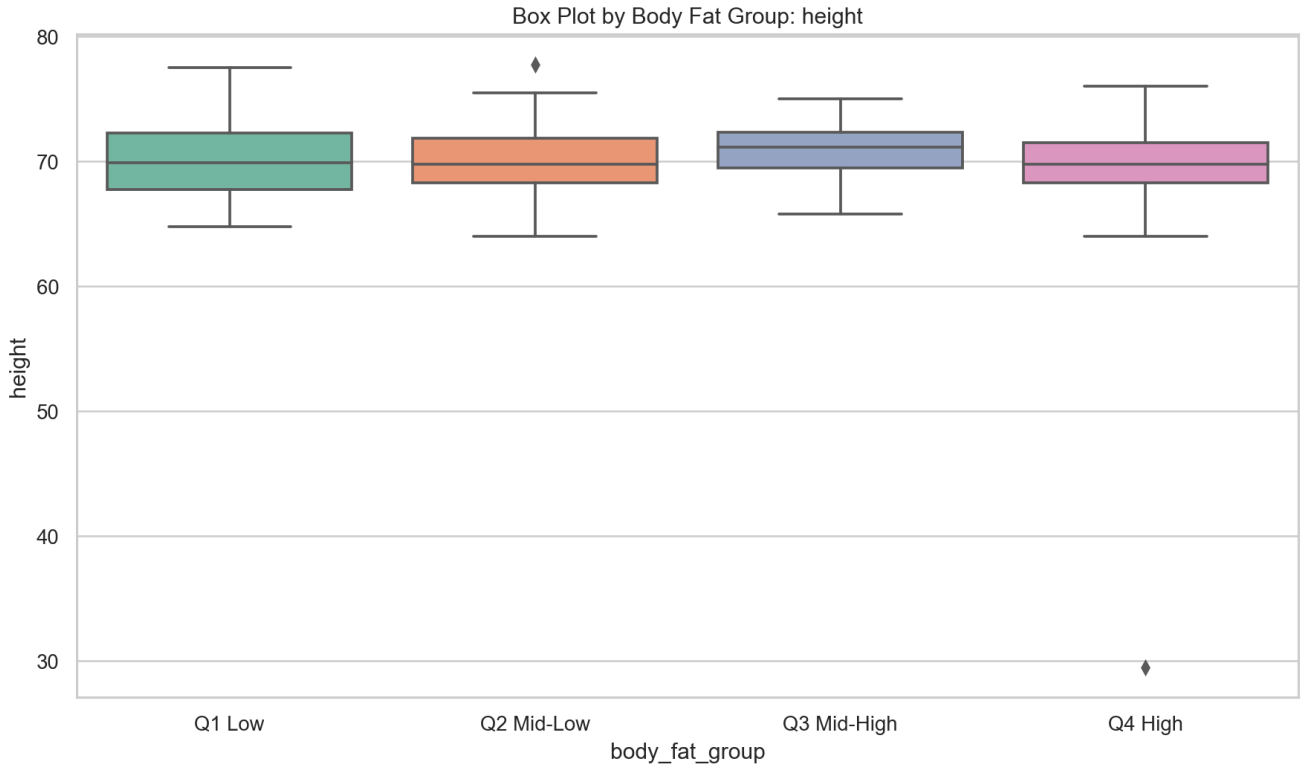
Grouped Box: tight



Grouped Box: neck



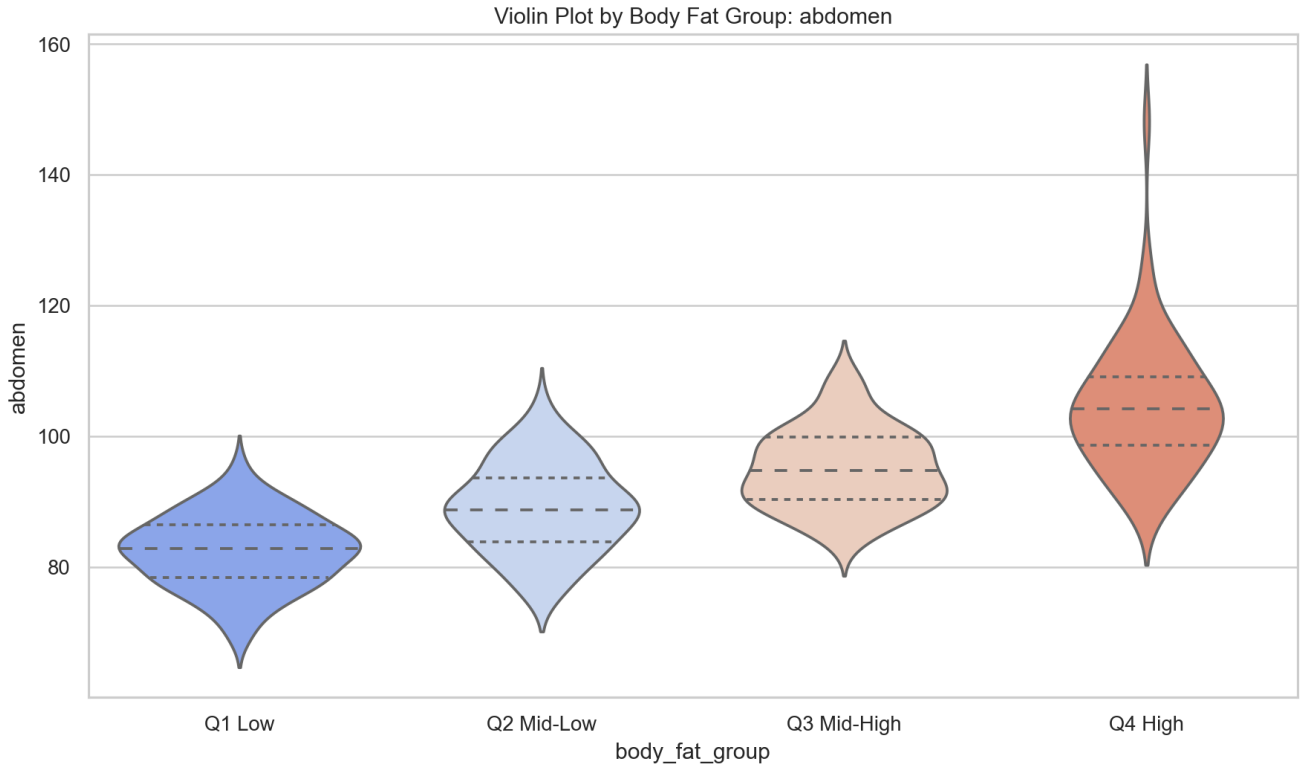
Grouped Box: height



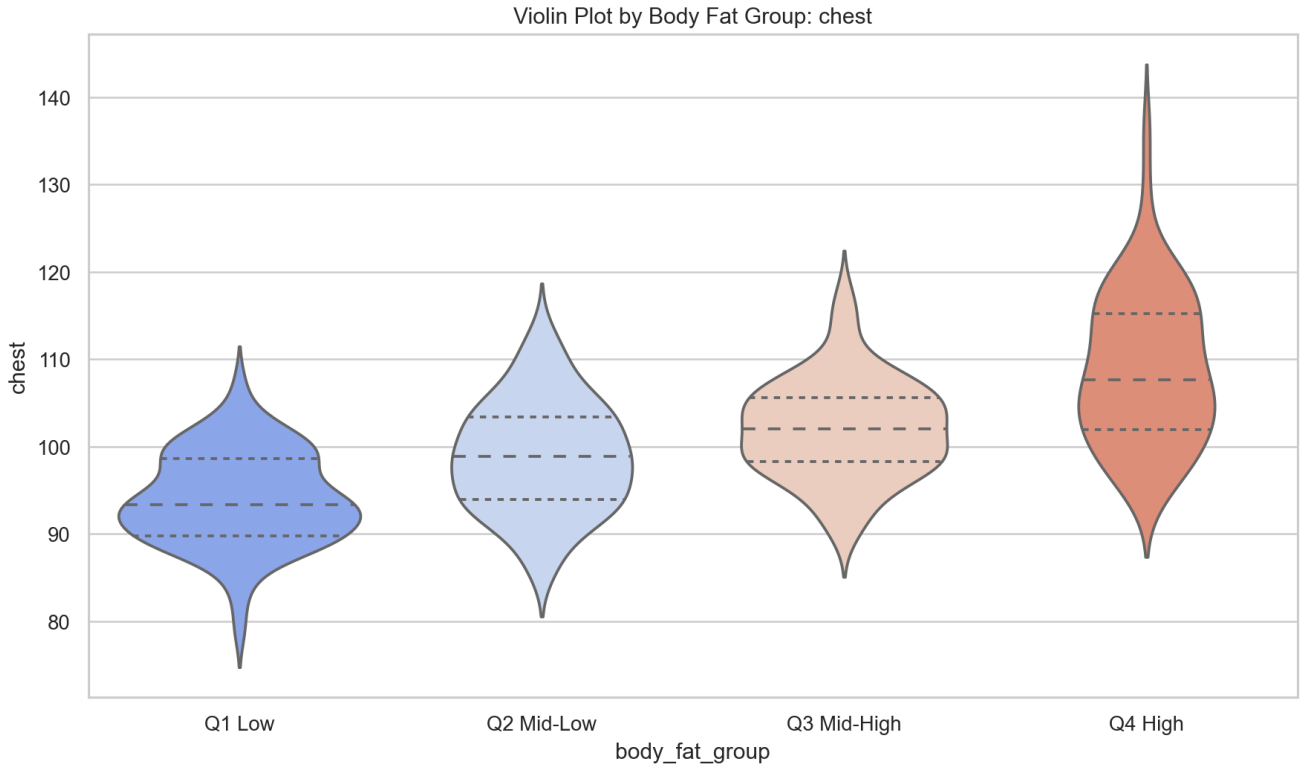
5) Violin Plot 해석 (분포 형태 + 밀도)

Violin plot으로 보면 그룹별 밀도 구조와 꼬리 길이를 확인할 수 있습니다. 일부 변수는 상위 그룹에서 분산이 커집니다.

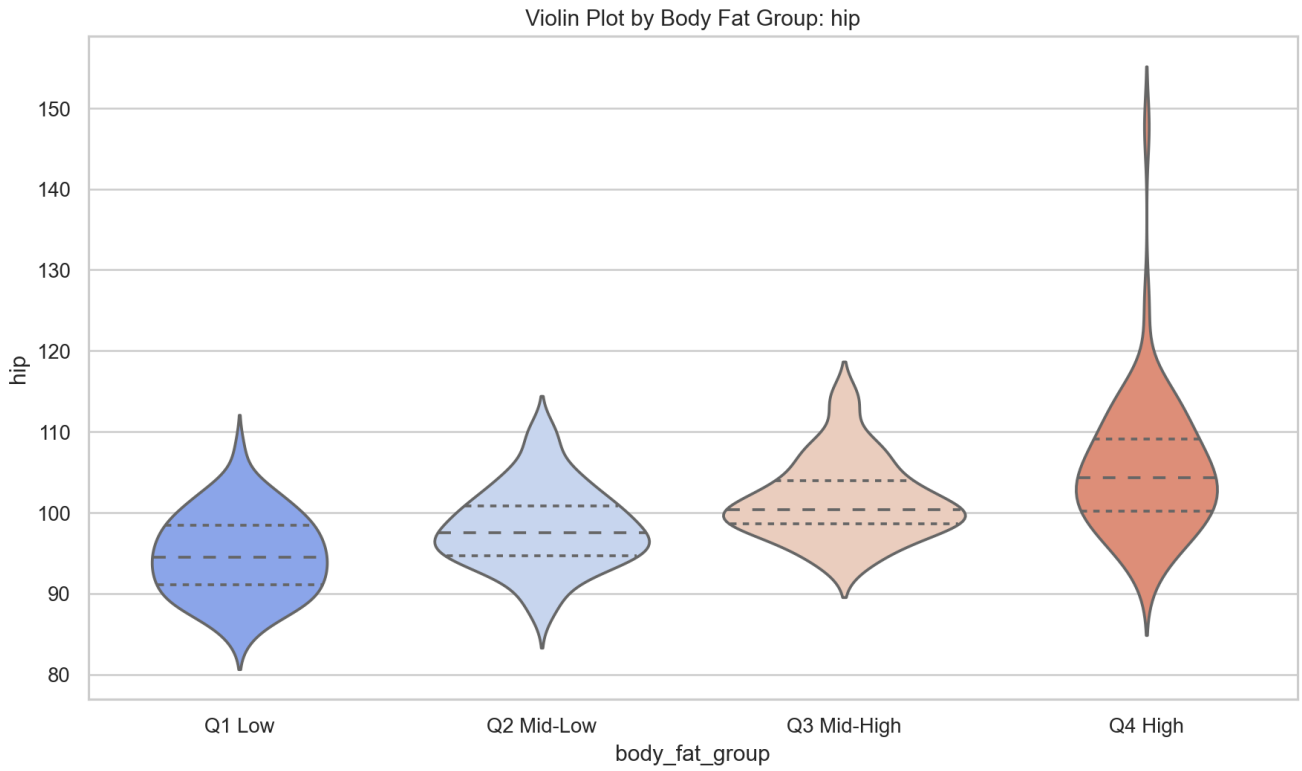
Grouped Violin: abdomen



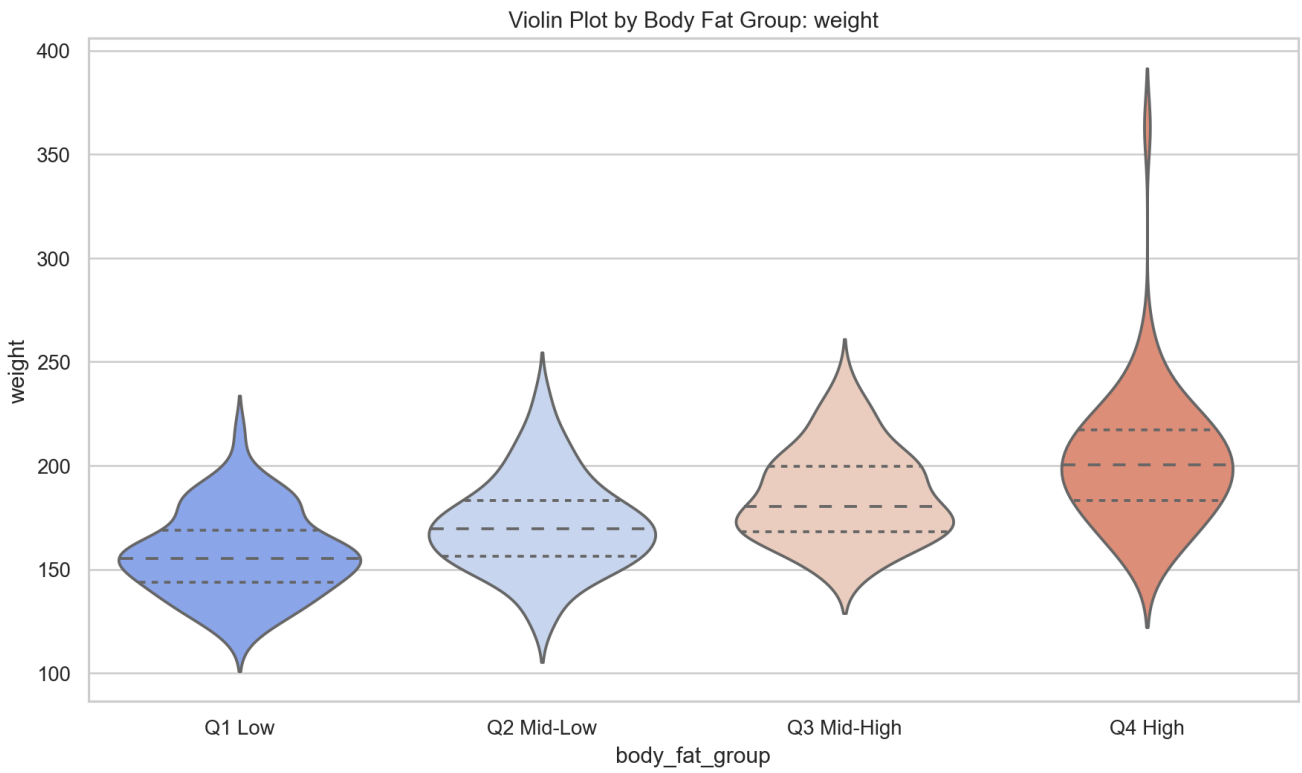
Grouped Violin: chest



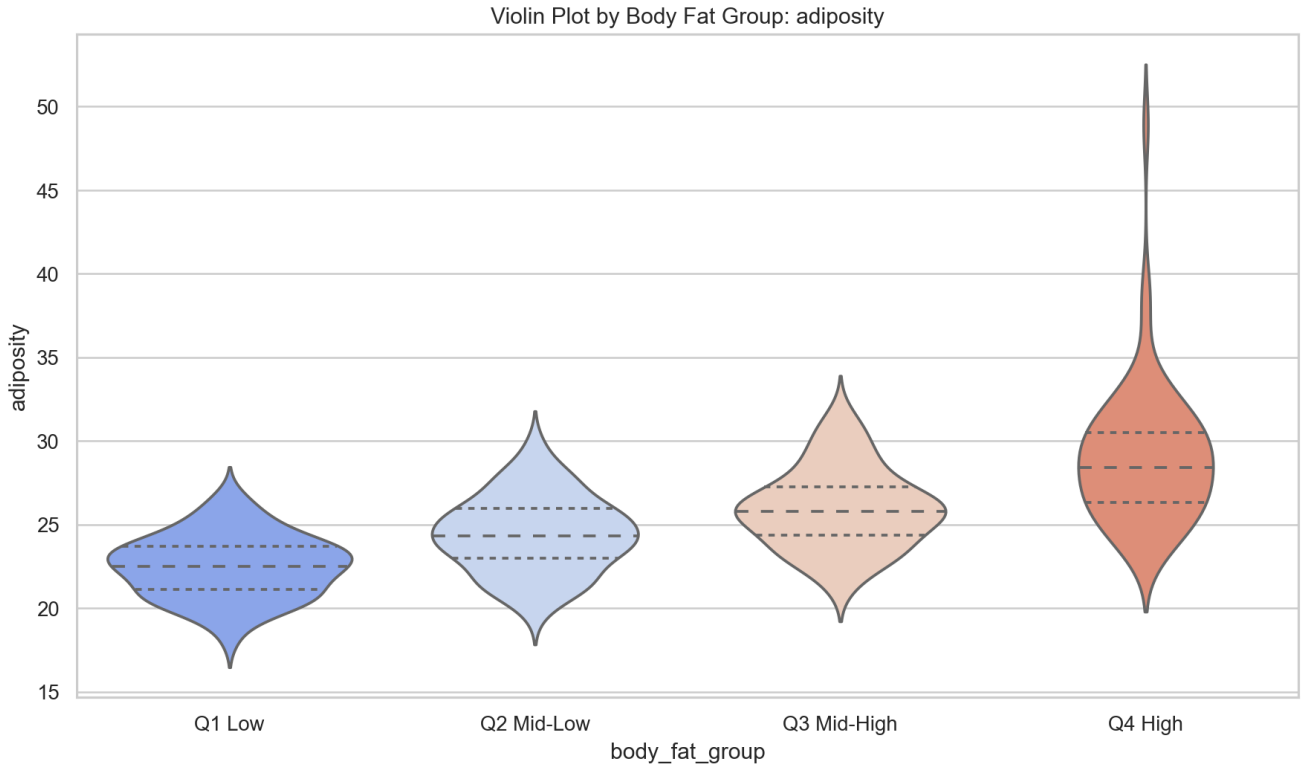
Grouped Violin: hip



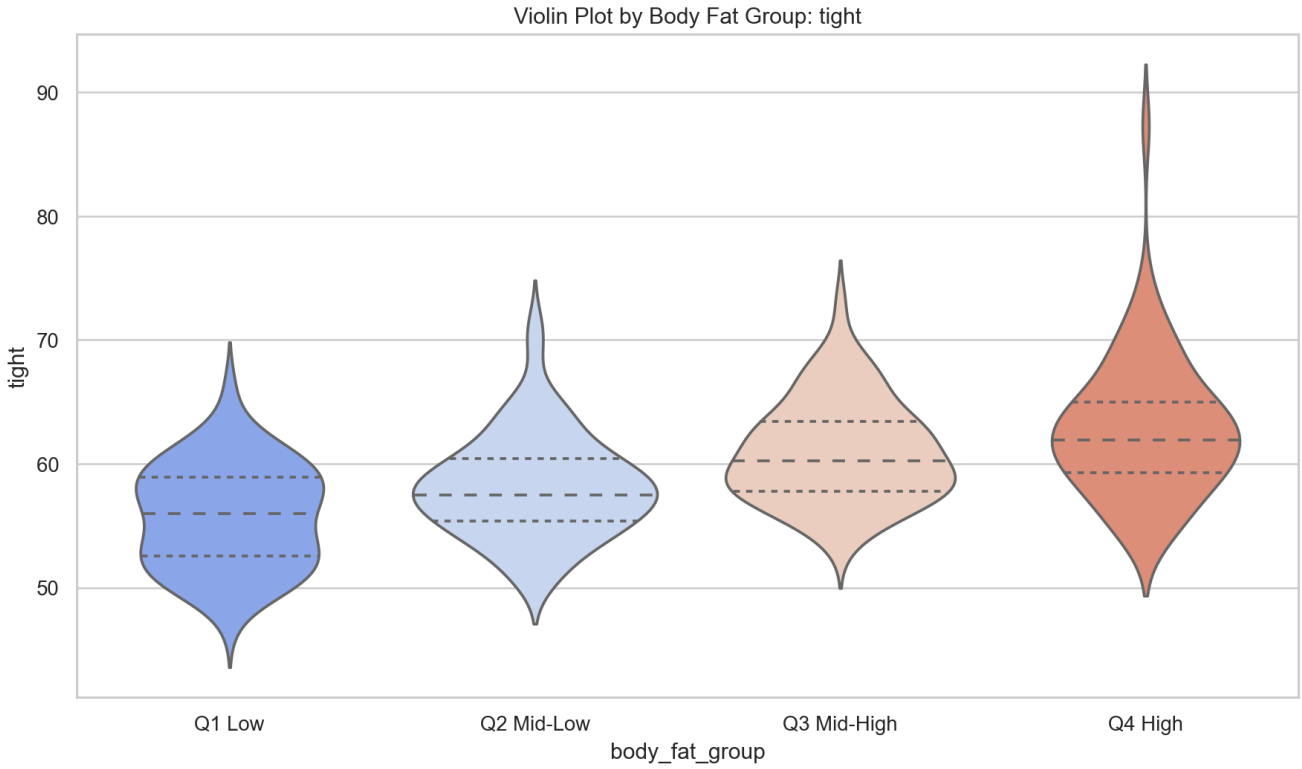
Grouped Violin: weight



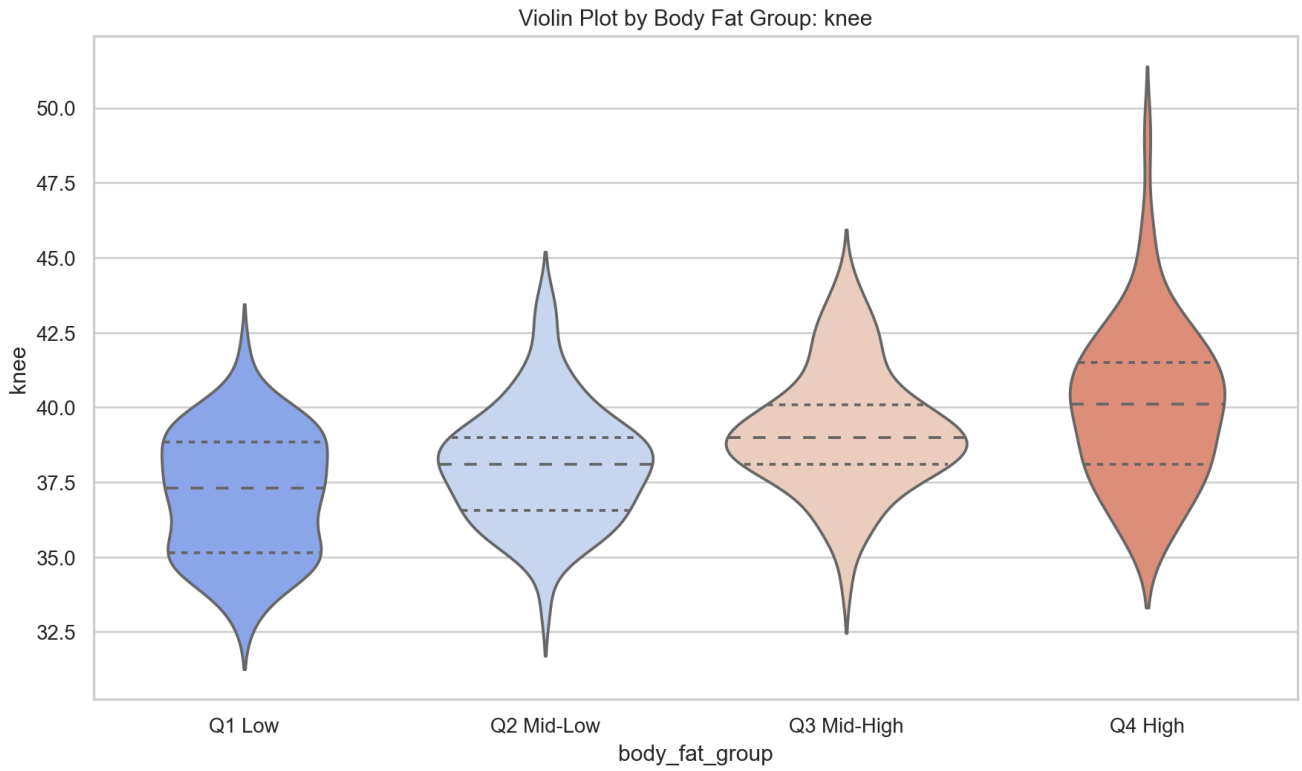
Grouped Violin: adiposity



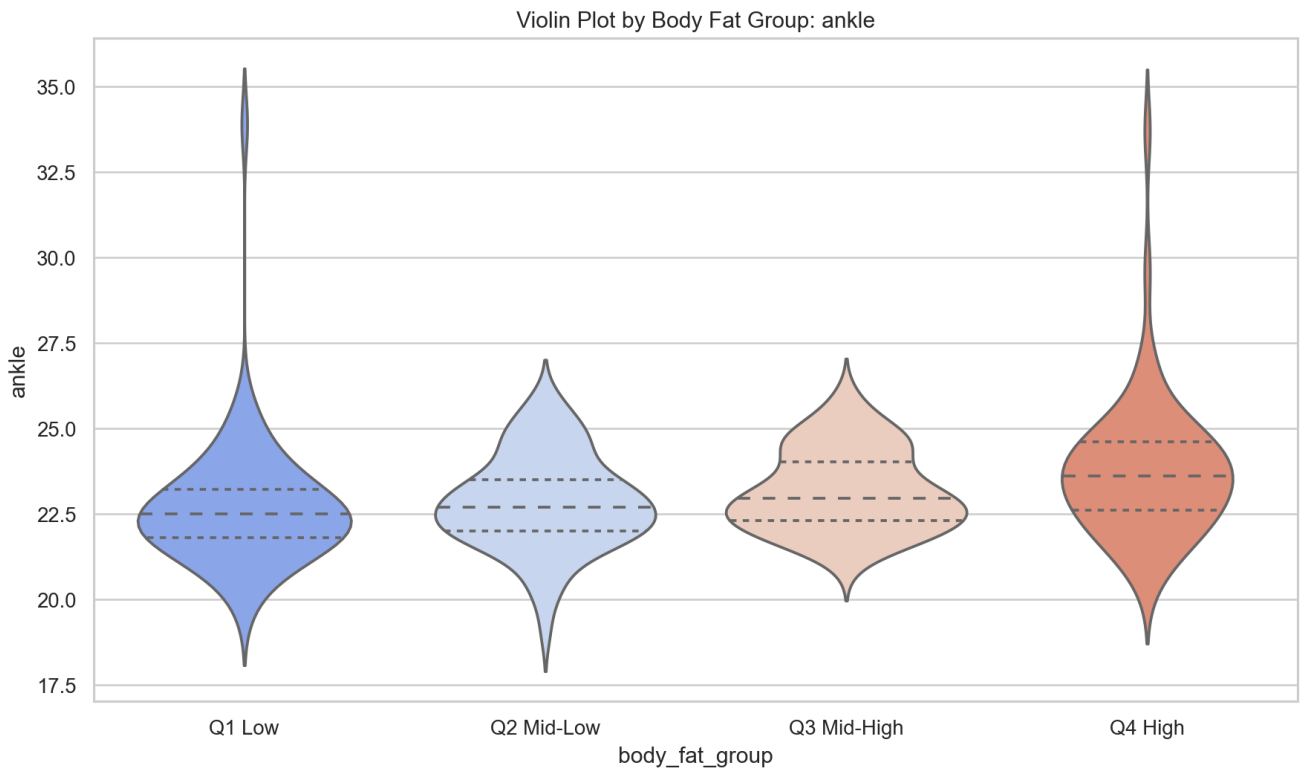
Grouped Violin: tight



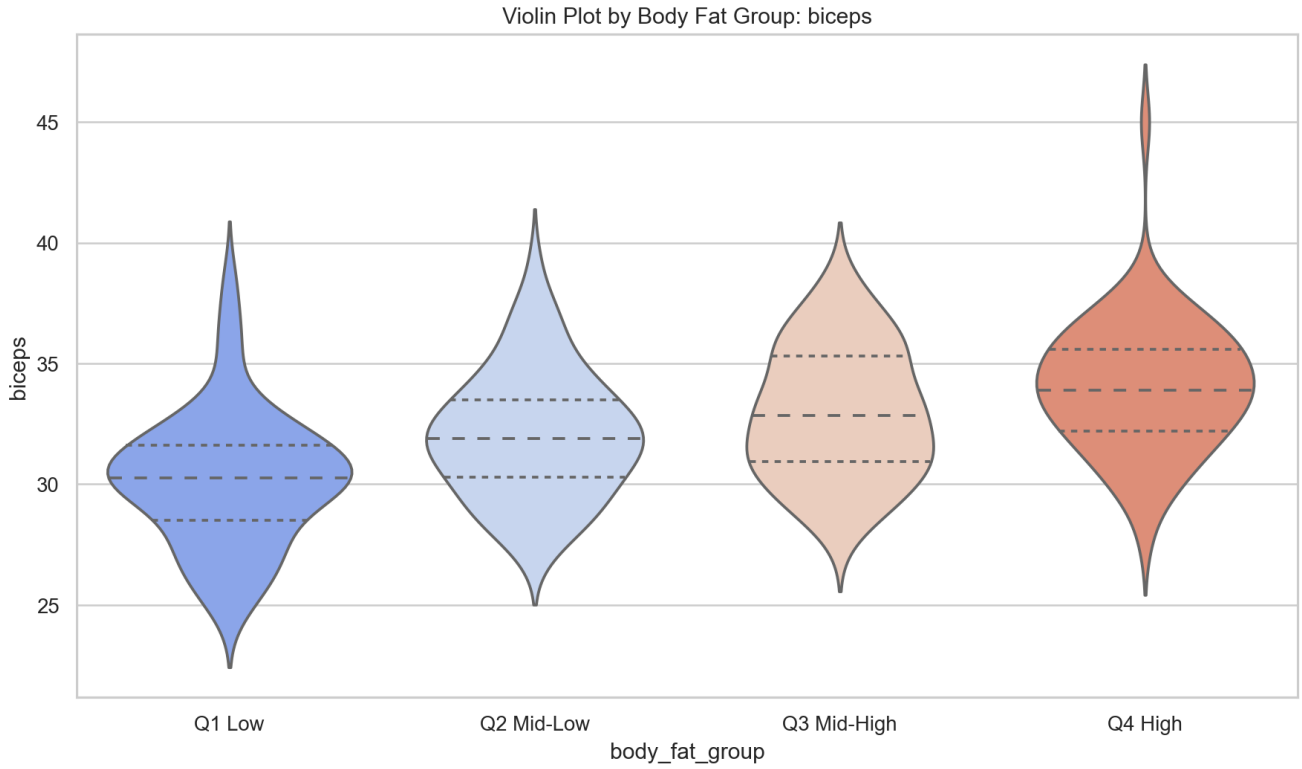
Grouped Violin: knee



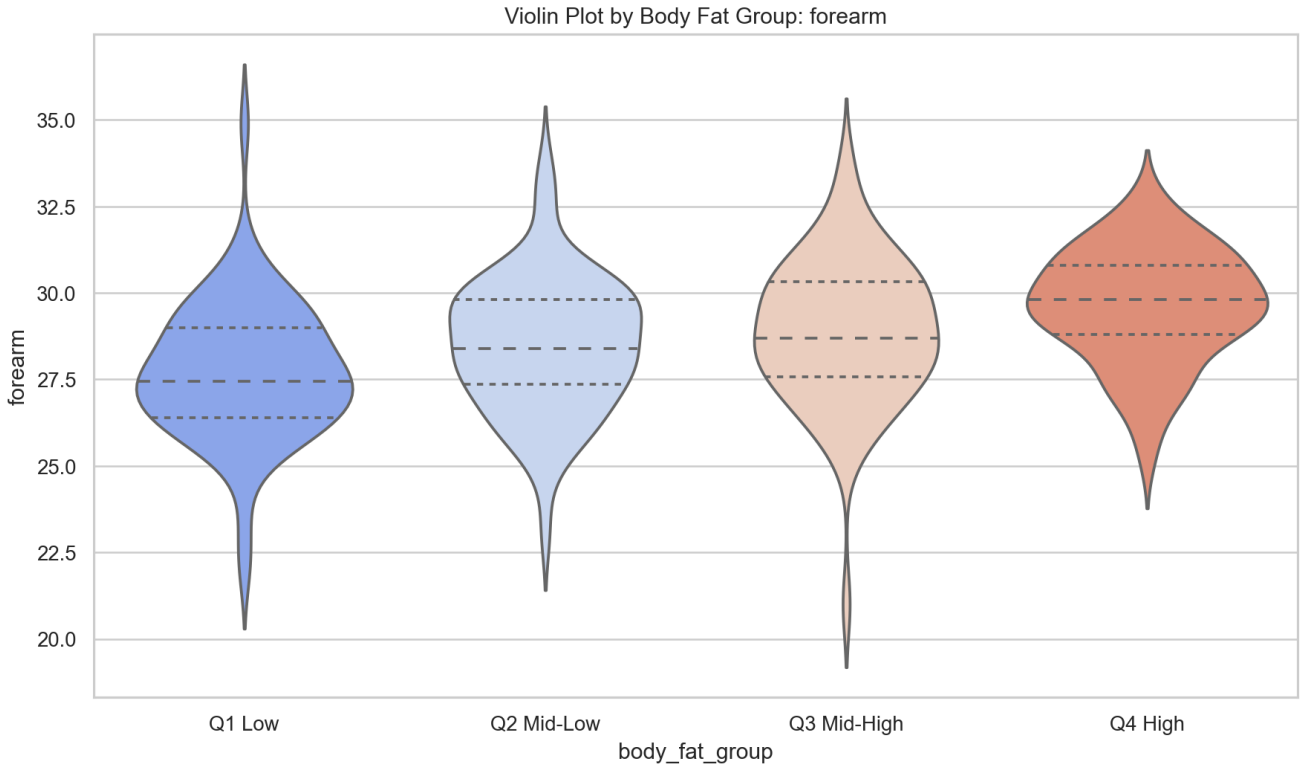
Grouped Violin: ankle



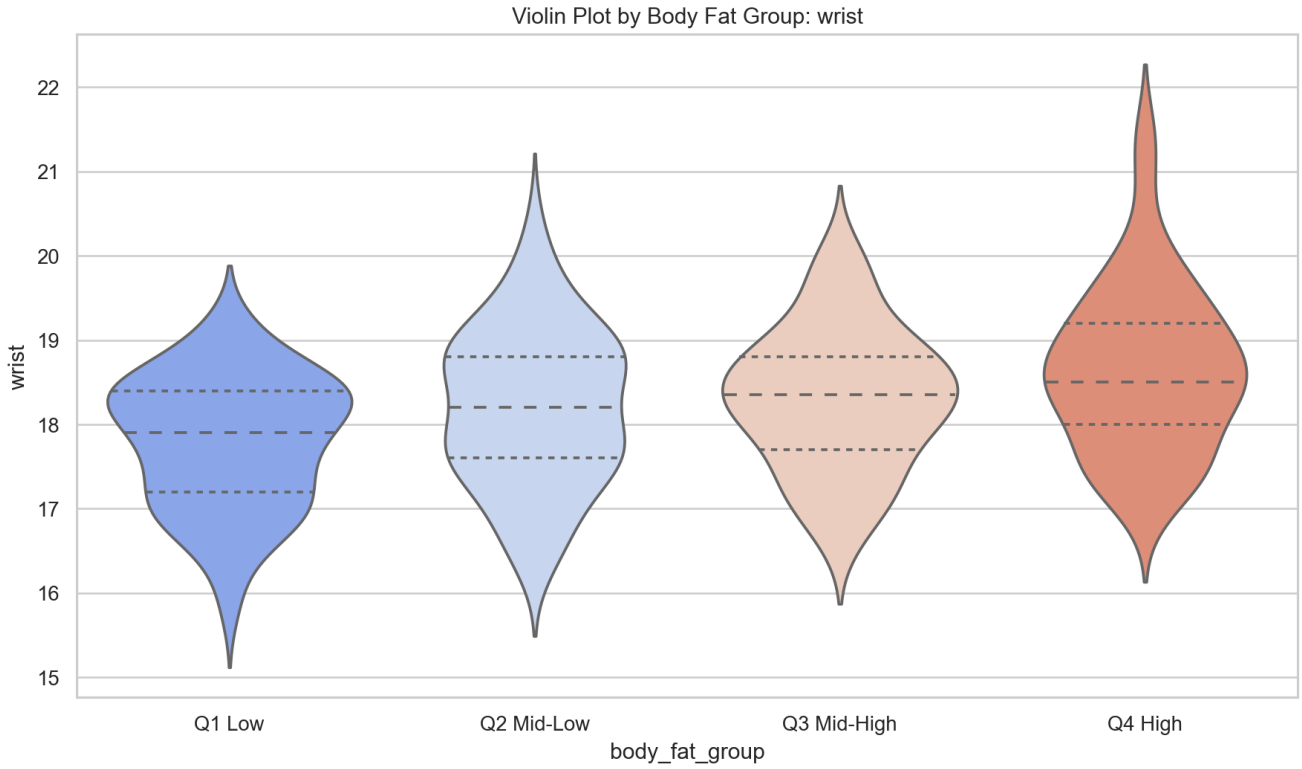
Grouped Violin: biceps



Grouped Violin: forearm



Grouped Violin: wrist



6) 종합 해석

- 복부 둘레(`abdomen`)가 체지방과 가장 강하게 연결되어, 실무적으로 가장 직관적인 설명 변수입니다.
- `adiposity`, `chest`, `hip`, `weight`도 체지방 증가와 함께 높아지는 경향이 강합니다.
- `height`는 상관이 약하고 음의 방향이라, 체지방 증가를 직접적으로 설명하는 힘은 제한적입니다.
- Box/Violin에서 Q4(고체지방) 그룹은 중심값 상승과 함께 분산 확장 경향이 나타납니다.
- 결론적으로 체지방 해석은 단일 변수보다 `abdomen + chest + adiposity + weight`의 조합으로 보는 것이 가장 설득력 있습니다.